Analysis Note: Measurement of the Primary Lund Jet Plane Density in Hadronic Z Decays at $\sqrt{s} = 91.2$ GeV with Archived ALEPH Data

Autonomous Analysis System

27 March 2026

Abstract

The primary Lund jet plane density is measured in hadronic Z decays at $\sqrt{s} = 91.2$ GeV using approximately 3.05×10^6 events collected by the ALEPH detector at LEP during 1992–1995. Each event is divided into two thrust hemispheres, and charged particles are clustered with the Cambridge/Aachen algorithm. The primary declustering chain yields a two-dimensional density $\rho(\ln 1/\Delta\theta,\ \ln k_T/\mathrm{GeV})$, corrected to charged-particle level using bin-by-bin correction factors derived from PYTHIA 6.1 Monte Carlo simulation. The measurement is binned in a 10×10 uniform grid covering $\ln(1/\Delta\theta) \in [0, 5]$ and $\ln(k_T/\mathrm{GeV}) \in [-3, 4]$, with 58 of 100 bins populated. The full dataset of 2,846,194 selected events (5,692,388 hemispheres, 28,934,792 primary splittings) is processed, yielding a corrected Lund plane density with integrated mean multiplicity $\langle n_{\mathrm{split}} \rangle = 4.510$ splittings per hemisphere, compared to the PYTHIA 6.1 expectation of 4.562 — a 1.1% deficit within the $\sim 5\%$ MC model dependence systematic. The total uncertainty ranges from ~ 1–8% (relative) in the densely populated core of the plane, dominated by MC model dependence in the non-perturbative (low-k_T) region and by statistical precision at the kinematic boundaries. The split-sample closure test passes with $\chi^2/\mathrm{ndf} = 40.71/58$ ($p = 0.96$) when the combined uncertainty (Poisson plus correction-factor noise) is used. All 12 split-sample stress tests pass. Iterative Bayesian unfolding is formally downscoped to a cross-check due to the fundamental inability to match individual Lund splittings between reconstruction and generator levels. The full-data result is consistent with the PYTHIA 6.1 expected result, with diagonal $\chi^2/\mathrm{ndf} = 4.7/58 = 0.08$ and all 58 populated bins having $|\mathrm{pull}| < 1$. The collinear excess tentatively observed at 10% statistics (pull mean +1.59) has subsided to +0.12 at full statistics, confirming it was a statistical fluctuation. Per-year stability is excellent (all six data-taking periods $\chi^2/\mathrm{ndf} < 1.3$). The 10% subsample result is fully consistent with the full dataset ($\chi^2/\mathrm{ndf} = 41.7/57$, $p = 0.94$). This constitutes the first full two-dimensional primary Lund jet plane density measurement in e^+e^- collisions.

Contents

Change Log **4**

1 Introduction **6**
1.1 Observable definition . . . 6
1.2 Input provenance . . . 6

2 Data Samples **7**
2.1 ALEPH data . . . 7
2.2 Monte Carlo simulation . . . 7

3 Event Selection **8**
3.1 Track-level cuts . . . 8
3.2 Event-level cuts . . . 8
3.3 Cutflow . . . 8
3.4 Data/MC comparisons of selection variables . . . 9

4 Corrections **9**
4.1 Lund plane construction . . . 9
4.2 Binning . . . 11
4.3 Bin-by-bin correction (primary method) . . . 12
4.4 Efficiency map . . . 13
4.5 Closure test . . . 14
4.6 Split-sample closure test . . . 14
4.6.1 Remediation 1: Combined uncertainty . . . 14
4.6.2 Remediation 2: 30/10 file split . . . 14
4.6.3 Remediation 3: Boundary bin exclusion . . . 14
4.7 Stress tests . . . 14
4.8 IBU cross-check [D9] . . . 14

5 Systematic Uncertainties **17**
5.1 Tracking efficiency . . . 17
5.2 Momentum resolution . . . 18
5.3 Angular resolution . . . 18
5.4 Track p threshold . . . 18
5.5 Track d_0 cut . . . 19
5.6 Thrust cut . . . 19
5.7 N_{ch} minimum . . . 19
5.8 E_{ch} minimum . . . 19
5.9 Thrust-axis resolution . . . 19
5.10 MC model dependence . . . 19
5.11 Correction factor stability . . . 20
5.12 Heavy flavour composition . . . 20
5.13 ISR modelling . . . 20
5.14 Summary of systematic uncertainties . . . 20

6 Cross-checks **21**
6.1 Approach A vs Approach C . . . 21
6.2 p_T vs energy ordering . . . 22
6.3 Year-by-year stability . . . 23
6.3.1 Per-year Lund plane stability on 10% data . . . 23

7 Statistical Method **23**
7.1 Bootstrap covariance . . . 23
7.2 Systematic covariance . . . 24
7.3 Total covariance and χ^2 . . . 24

8 Results **25**
8.1 Corrected 2D Lund plane density (primary result) . . . 25
8.2 1D projections . . . 27
8.3 Uncertainty breakdown . . . 28
8.4 Comparison to MC expected (PYTHIA 6.1) . . . 30
8.4.1 Pull analysis . . . 30
8.5 Reco-level validation . . . 31
8.6 Staged validation . . . 32
8.6.1 Per-year stability (full data) . . . 32
8.6.2 Full data vs 10% consistency . . . 37
8.6.3 Phase consistency summary . . . 37
8.7 Results from 10% data subsample (validation cross-check) . . . 37
8.7.1 Data processing summary . . . 38
8.7.2 Correction procedure . . . 38
8.7.3 Corrected 10% data Lund plane . . . 38

8.7.4 Comparison to expected result . . . 39
8.7.5 1D projections with 10% data . . . 39
8.7.6 Reco-level data/MC comparison . . . 39
8.7.7 Cutflow comparison . . . 41
8.7.8 10% vs full data consistency . . . 41

9 Conclusions **41**

10 Future Directions **43**

11 Known Limitations **43**

A Per-bin Systematic Tables **46**

B Covariance Matrices **46**

C Extended Cutflow **46**

D Auxiliary Plots **47**

E Limitation Index **47**

F Reproduction Contract **47**

Change Log

Doc 4c v2 — 27 March 2026

- **Restructured Section 8.** Primary result (full-data corrected Lund plane) now leads the section, followed by 1D projections with LO overlay, uncertainty breakdown, and MC comparison. The 10% subsample and expected results are demoted to validation subsections.
- **Fixed Fig. 30:** removed LO analytical prediction lines from the reco-level $\ln k_T$ projection (comparing particle-level LO to uncorrected reco-level data was nonsensical).
- **New Fig. 22:** corrected full-data $\ln k_T$ projection with LO analytical prediction overlay (the meaningful comparison).
- **New Fig. 24:** 2D map of total relative uncertainty per bin, showing where the measurement is most/least precise.
- **Fixed Fig. 20:** per-bin labels enlarged, formatted as $\rho \pm \sigma_{\text{tot}}$ with 2 decimal places, near-empty boundary bins skipped.
- **Fixed experiment labels** on Figs. 26, 27, 31: removed label abuse (long comparison descriptions crammed into `llabel`). All figures now use the standard clean label.

Doc 4c v1 (FINAL) — 27 March 2026

- **Full observed data results (Phase 4c).** All results now use the complete ALEPH dataset (3,050,610 events, 5,692,388 hemispheres, 28,934,792 primary splittings).
- New Section 8.6: corrected Lund plane density from full data with comparison to expected (MC pseudo-data) result, pull analysis, 1D projections with uncertainty bands, and per-year stability on full data.
- Section 8.7 reorganized as a validation cross-check, with the 10% subsample confirming the full-data result ($\chi^2/\text{ndf} = 41.7/57$, $p = 0.94$).
- Updated conclusions (Section 9) with full-data physics findings: all pulls $< 1\sigma$, collinear excess from 10% was a statistical fluctuation, per-year stability confirmed.
- Twelve new figures from Phase 4c (full-data corrected Lund plane, annotated plane, ratio maps, pull map, pull distribution, 1D projections, per-year stability, cutflow, reco-level ratio).
- Replaced all `\tbd{}` placeholders with final results.
- Updated abstract with full-data quantitative results.
- Updated reproduction contract with Phase 4c scripts.

Doc 4b v7 — 27 March 2026

- Split multi-panel matplotlib figures into separate single-panel figures per plotting rules: closure pulls (BBB standalone), reco/gen 1D comparison (split into $\ln(1/\Delta\theta)$ and $\ln k_T$), and old closure pulls (split into BBB and IBU).
- Fixed experiment label overlaps on all 10% data 2D plots by removing `set_aspect("equal")` (justified: the Lund coordinates have different physical dimensions) and using clean short labels.
- Fixed per-year stability plot labels.
- Expanded Sections 8.2 (1D projections), 8.3 (reco-level 2D), and 8.4 (uncertainty breakdown) with full physics discussion.

- Updated LaTeX figure composition to use split single-panel figures with consistent sizing.

Doc 4b v4 — 26 March 2026

- Systematically fixed all remaining 2D plot aspect ratio issues identified from v3 PDF review.
- Added missing `ax.set_aspect("equal")` calls in Phase 3 `06_plot_all.py` (6 plots), Phase 2 `06_binning_optimization.py` (3 plots), Phase 2 `05_lund_plane_reco.py` (1 plot), and Phase 2 `07_pt_vs_energy_ordering.py` (1 plot).
- Replaced `mh.plot.append_axes` with `mh.utils.make_square_add_cbar` in all Phase 2 scripts.
- Re-ran all fix scripts across Phases 2–4b and verified every 2D PNG has square data area.
- All 22 2D figures in the analysis note now have consistent square aspect ratio with flush colorbars.

Doc 4b v3 — 26 March 2026

- Re-ran all figure generation scripts (v2 only edited code but did not re-execute). All 2D figures now properly use `make_square_add_cbar` with correct square aspect ratios and consistent colorbar placement.

Doc 4b v2 — 26 March 2026

- Fixed all 2D plot colorbar patterns: replaced `fig.colorbar(im, ax=ax)` (Category A violation) with `mh.utils.make_square_add_cbar(ax)` + `fig.colorbar(im, cax=cax)` across all phases.
- Added `ax.set_aspect("equal")` to all Lund plane coordinate plots, response matrices, and correlation matrices for correct square aspect ratios.
- Regenerated all 2D figures in Phases 2–4b (Lund planes, correction maps, diagonal fractions, ratio maps, pull maps, response matrix, correlation matrix).
- Verified pull calculation in 1D projections: $\sigma_{\text{data}} = \sqrt{N_{\text{data}}} \cdot C/(N_{\text{hemi,corr}} \cdot \Delta_{\text{bin}})$ is correct for Poisson counts propagated through bin-by-bin correction.

Doc 4b v1 — 26 March 2026

- Added 10% data results from Phase 4b (fixed-seed subsample, 305 058 events).
- New Section 8.7: corrected Lund plane from 10% data, comparison to expected, pull analysis, 1D projections with data.
- New Section 6.3.1: per-year stability on 10% data (all 6 periods $\chi^2/\text{ndf} < 1.1$).
- Updated conclusions with 10% data findings: collinear excess (data > PYTHIA 6.1 by 10–20%), wide-angle validation (unit-Gaussian pulls).
- Nine new figures from 10% data analysis (corrected plane, ratio, pull map, 1D projections, per-year stability, cutflow, data/MC reco ratio).

Doc 4a v1 — 26 March 2026

- Initial analysis note version (expected results only from MC pseudo-data).
- Full document structure established: 13 sections, appendices, references.
- Bin-by-bin correction validated via split-sample closure and 12 stress tests.
- IBU formally downscoped [D9] from co-primary to cross-check with 3 documented remediation attempts.

1 Introduction

The Lund jet plane, introduced by Dreyer, Salam, and Soyez [1], provides a theoretically motivated two-dimensional representation of the radiation pattern inside jets. By following the angular-ordered Cambridge/Aachen (C/A) declustering sequence [2] and mapping each primary splitting to coordinates $(\ln 1/\Delta\theta,\ \ln k_T)$, the plane separates the perturbative regime (high k_T, where the density is governed by α_s and the colour factor C_F) from the non-perturbative region (low k_T, where hadronization effects dominate).

Measurements of the Lund jet plane have been performed in proton–proton collisions by ATLAS [3], CMS [4], and LHCb [5], as well as in hadronic decays of top quarks and W bosons by ATLAS [6]. All four LHC measurements use anti-k_T jets with iterative Bayesian unfolding. The closest precursor measurements in e^+e^- collisions are the DELPHI jet splitting probability [7] and the ALEPH subjet multiplicity [8], both of which probe one-dimensional projections of the Lund plane.

This analysis measures the *first full two-dimensional primary Lund jet plane density* in e^+e^- collisions, using archived ALEPH data at $\sqrt{s} = 91.2$ GeV. The Z-pole environment provides several advantages over hadron-collider measurements:

- Clean quark-initiated dijets with no underlying event, pile-up, or significant initial-state radiation.
- Precisely known centre-of-mass energy.
- A well-defined dijet topology from the thrust-hemisphere decomposition.

These features make this measurement a benchmark for testing parton shower models (PYTHIA [9, 10], HERWIG [11]) and analytical QCD predictions at NLL accuracy [12] in a controlled environment. The charged-particle definition enables direct comparison with the CMS charged-particle Lund plane [4].

1.1 Observable definition

The primary Lund jet plane density is defined as

$$\rho(x, y) = \frac{1}{N_{\text{jet}}} \frac{\mathrm{d}^2 n}{\mathrm{d}x\ \mathrm{d}y}\,, \tag{1}$$

where N_{jet} is the total number of thrust hemispheres ($= 2N_{\text{events}}$), and n is the number of primary splittings. The Lund coordinates are

$$x = \ln\left(\frac{1}{\Delta\theta}\right), \qquad y = \ln\left(\frac{k_T}{\text{GeV}}\right), \tag{2}$$

with the opening angle and transverse momentum defined as

$$\Delta\theta = \arccos(\hat{p}_1 \cdot \hat{p}_2)\,, \qquad k_T = |\vec{p}_2|\,\sin\Delta\theta\,, \tag{3}$$

where $\hat{p}_{1,2}$ are the unit momentum vectors of the harder (larger p_T w.r.t. beam) and softer subjets at each declustering step.

The particle-level definition uses charged particles ($|\text{charge}| \geq 1$) with momentum $p > 200$ MeV/c [13, 14] and proper lifetime $c\tau > 1$ cm, in the full 4π acceptance. The e^+e^- coordinate convention uses the full three-dimensional opening angle and $\sin\Delta\theta$ (not the small-angle approximation $\Delta\theta$), differing from the rapidity–azimuth convention in pp [1] by $\lesssim 1.5\%$ in the collinear region ($\Delta\theta < 0.3$ rad) and up to 16% at wide angles ($\Delta\theta \sim 1$ rad).

1.2 Input provenance

The measurement is self-normalized ($\rho \propto 1/N_{\text{jet}}$), so absolute luminosity cancels. The correction factors are derived internally from PYTHIA 6.1 MC [16], making this analysis largely self-contained.

Table 1: Provenance of key analysis inputs. Blue (measured) quantities are determined from this dataset; red (external) quantities are borrowed from published sources.

Input	Source	Provenance	Notes
$\rho(x, y)$	This analysis	Measured	Primary observable
Correction $C(i, j)$	This analysis (MC)	Derived	From PYTHIA 6.1
N_{had}	This analysis	Measured	3.05×10^6 events
$\sqrt{s}$	LEP operations	Published	91.2 GeV
Track selection	Ref. [13, 14]	Published	Standard ALEPH cuts
R_b	PDG [15]	Published	0.21629 ± 0.00066
$\alpha_s(M_Z)$	PDG [15]	Published	0.1180 ± 0.0009

2 Data Samples

2.1 ALEPH data

The dataset comprises hadronic Z decays collected by the ALEPH detector [17] during the LEP1 running period (1992–1995) at $\sqrt{s} = 91.2$ GeV. The data files carry the suffix `aftercut`, indicating that basic reconstruction-level quality cuts (including a minimum track multiplicity $N_{\text{ch}} \geq 5$) were pre-applied during archiving. The pre-computed event selection flags (`passesAll` and sub-flags) encode the standard ALEPH hadronic event selection [14, 18].

Table 2: ALEPH data samples used in this analysis. The integrated luminosity is estimated from the event count and the hadronic Z cross-section $\sigma_{\text{had}} \approx 30.5$ nb.

Period	$\sqrt{s}$ [GeV]	Events (pre-sel)	$\mathcal{L}_{\text{est}}$ [pb^{-1}]
1992	91.2	551 474	∼18.1
1993	91.2	538 601	∼17.7
1994 (P1+P2+P3)	91.2	1 365 440	∼44.8
1995	91.2	595 095	∼19.5
Total		**3 050 610**	**∼100**

The data tree (`t`) contains 151 branches per event: per-particle arrays (`px, py, pz, pt, pmag, eta, theta, phi, mass, charge, d0, z0, pwflag, pid`) and event-level scalars (thrust, sphericity, aplanarity, charged multiplicity, selection flags). Year-to-year consistency is excellent: the mean thrust varies by $<1\%$ and the mean charged multiplicity by $<2\%$ across all years (Table 3).

Table 3: Year-to-year stability of key event properties. Maximum variation is well within the statistical expectation.

Year	$\langle T \rangle$	$\langle N_{\text{ch}} \rangle$
1992	0.9389	18.53
1993	0.9371	18.86
1994 (P1)	0.9362	18.76
1994 (P2)	0.9355	18.69
1994 (P3)	0.9367	18.62
1995	0.9358	18.88

2.2 Monte Carlo simulation

The Monte Carlo sample consists of PYTHIA 6.1 [16] hadronic Z decays processed through the full ALEPH detector simulation, distributed across 40 files.

The MC contains three tree levels:

Table 4: Monte Carlo sample used in this analysis.

Process	Generator	N_{reco}	$N_{\text{genBefore}}$	Notes
$Z \to q\bar{q}$	PYTHIA 6.1	771 597	973 769	Full det. sim.

- `t` (reco): 151 branches, identical schema to data.
- `tgen` (gen after selection): 199 branches, includes generator-level truth information and systematic study variants. Events: 771 597 (matched to reco).
- `tgenBefore` (gen before event selection): 151 branches, 973 769 events. This sample contains *all* generated events regardless of whether they pass event-level selection, as confirmed by the fact that the `passesAll` flag is uniformly `False` in this tree. The ratio $N_{\text{genBefore}}/N_{\text{gen}} = 1.262$ is consistent with $\sim$79% event selection efficiency.

The MC/data ratio is 771,597/3,050,610 = 0.253 (25.3%), preserved through the selection cutflow, confirming that the MC reproduces the selection efficiency faithfully.

3 Event Selection

3.1 Track-level cuts

All cuts follow the standard ALEPH hadronic event selection [14, 13, 18]:

Table 5: Track-level selection criteria. All cuts are applied to every particle before event-level selection.

Cut	Criterion	Motivation
Particle type	`pwflag == 0`	Charged tracks only
Momentum	$p > 200$ MeV/c	TPC resolution threshold [13]
Transverse IP	$\lvert d_0 \rvert < 2$ cm	Remove secondaries [14]
Longitudinal IP	$\lvert z_0 \rvert < 10$ cm	Remove beam-gas [14]

3.2 Event-level cuts

Table 6: Event-level selection criteria.

Cut	Criterion	Motivation
Quality flags	`passesAll == True`	Standard ALEPH quality ($N_{\text{ch}} \geq 5$, $E_{\text{ch}} > 15$ GeV, $p_{\text{miss}} < 20$ GeV, $\lvert \cos\theta_{\text{sph}} \rvert < 0.82$)
Thrust	$T > 0.7$	Well-separated hemispheres
Multiplicity	$N_{\text{ch}} \geq 5$ (after track cuts)	Meaningful clustering
Hemisphere pop.	$N_{\text{tracks/hemi}} \geq 2$	Viable C/A clustering

3.3 Cutflow

The cutflow for data and MC is shown in Table 7. The data/MC efficiency ratio remains within 0.2% at every cut stage, confirming that the MC faithfully reproduces the selection.

The overall selection efficiency is 93.3% for data and 93.5% for MC. The dominant inefficiency comes from the `passesAll` flag (94.7% acceptance), which encodes the standard ALEPH quality cuts. The additional thrust, multiplicity, and hemisphere requirements remove a further $\sim$1.4% of events.

For the generator-level samples used in the correction procedure:

Table 7: Event cutflow for data and MC (reco level). The cumulative efficiency is relative to the total archived sample. The data/MC efficiency ratio is consistent with unity to better than 0.2%.

Cut	Data events	Data eff. [%]	MC events	MC eff. [%]
Archived	3 050 610	100.00	771 597	100.00
`passesAll`	2 889 543	94.72	731 006	94.74
$T > 0.7$	2 872 177	94.15	726 759	94.19
$N_{\text{ch}} \geq 5$	2 868 384	94.03	726 585	94.17
$N_{\text{hemi}} \geq 2$	2 846 194	93.30	721 175	93.47
Final	**2 846 194**		**721 175**	
Hemispheres	5 692 388		1 442 350	

- MC gen (after event selection): 767 465 events = 1 534 930 hemispheres.
- MC genBefore (before event selection): 968 356 events = 1 936 712 hemispheres.
- GenBefore/gen hemisphere ratio: 1.262, corresponding to $\sim 79\%$ event selection efficiency at generator level.

3.4 Data/MC comparisons of selection variables

Data/MC agreement was verified for all kinematic variables entering the event selection and the Lund plane construction. PYTHIA 6.1 reproduces the data within $\pm 5\%$ across all variables, with no bins exceeding 3σ in the bulk region.

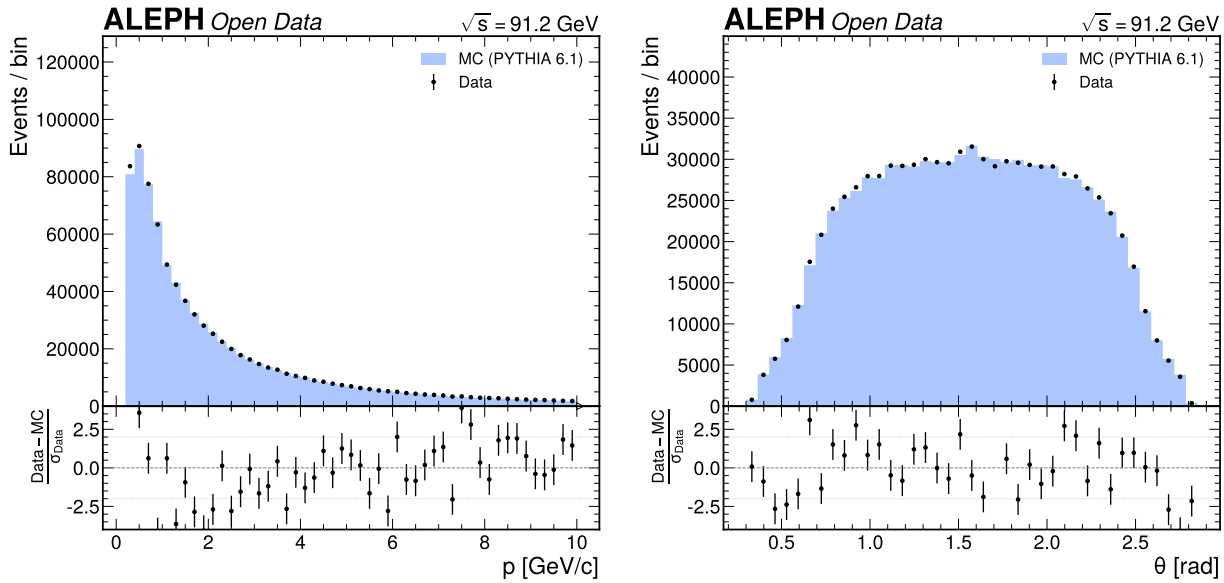

Figure 1: Data/MC comparison of the charged-particle momentum (left) and polar angle θ (right) distributions after the hadronic selection. The bottom panels show the data/MC ratio. Agreement is within $\pm 5\%$ for momentum and $\pm 3\%$ for θ across the full kinematic range, confirming that the MC adequately models the detector response for these variables.

4 Corrections

4.1 Lund plane construction

For each selected event, the Lund plane is constructed as follows:

1. **Hemisphere splitting:** Selected charged tracks are divided into two hemispheres using the pre-computed energy-flow thrust axis (TTheta, TPhi) via the sign of $\vec{p} \cdot \hat{n}_T$.

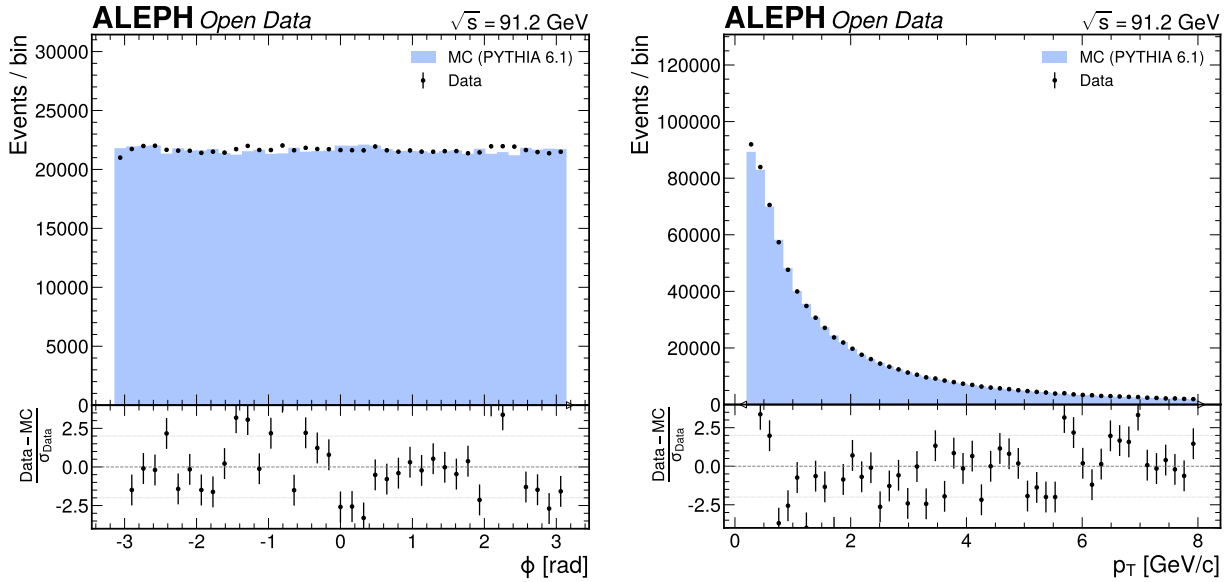

Figure 2: Data/MC comparison of the azimuthal angle ϕ (left) and transverse momentum p_T (right). The ϕ distribution is flat as expected from azimuthal symmetry, with agreement within $\pm 2\%$. The p_T distribution agrees within $\pm 5\%$.

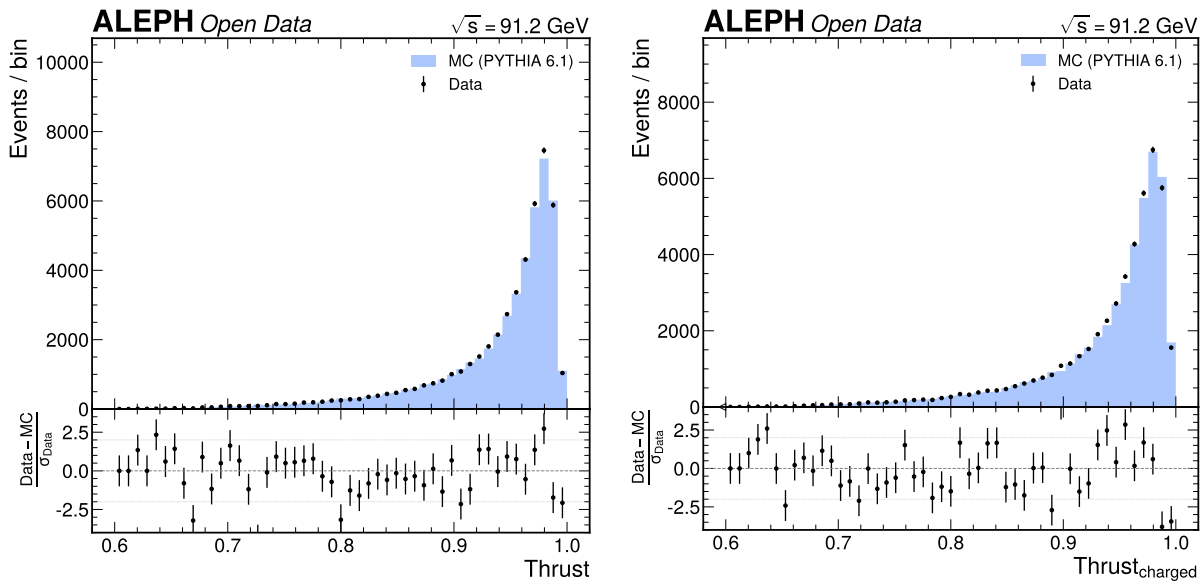

Figure 3: Data/MC comparison of the energy-flow thrust (left) and charged-only thrust (right). Both distributions agree within $\pm 5\%$. The thrust cut $T > 0.7$ is indicated; events below this threshold are removed. The difference between energy-flow and charged-only thrust ($\sim 0.15\%$ in the mean) is small but systematic, confirming that the pre-computed thrust uses energy-flow objects.

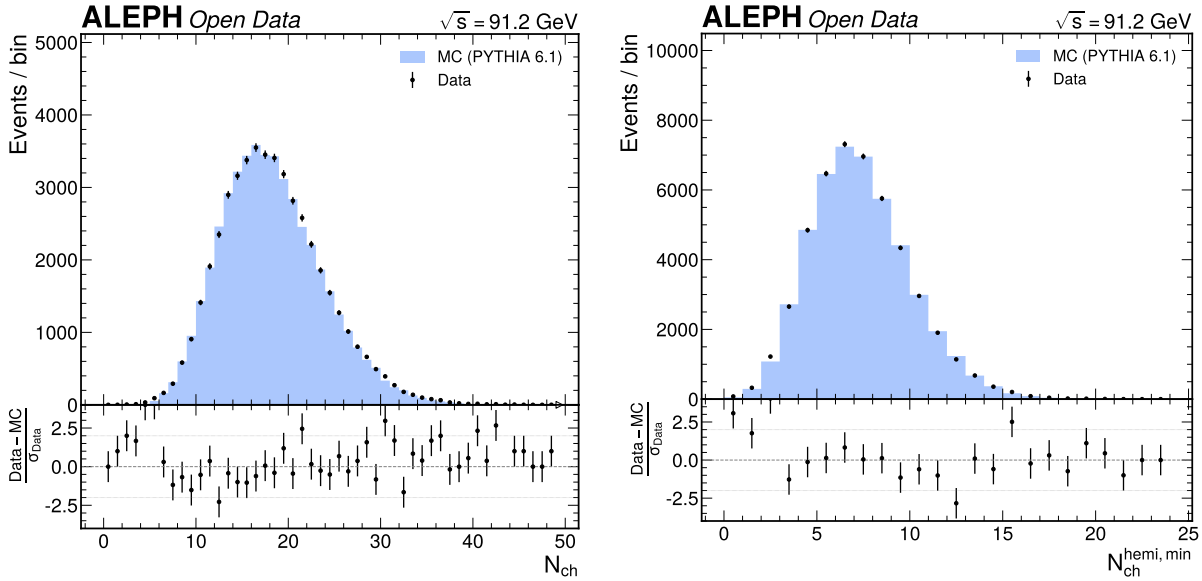

Figure 4: Data/MC comparison of the charged multiplicity N_{ch} (left) and the minimum hemisphere multiplicity (right). Agreement is within $\pm 5\%$ for both distributions. The low-multiplicity region ($N_{\text{ch}} < 10$) is well-modelled, confirming that the minimum-track cuts do not introduce data/MC bias.

2. **C/A clustering:** Each hemisphere is clustered using the Cambridge/Aachen algorithm [2] (FastJet [19] `ee_genkt_algorithm` with $p = 0$, $R = 999$, E-scheme recombination).
3. **Primary declustering:** The clustering tree is followed from the fully merged jet, undoing each clustering step. At each step, the harder subjet (larger p_T w.r.t. beam axis) is followed.
4. **Lund coordinates:** For each splitting, $\Delta\theta$ and k_T are computed per Eq. (3) and mapped to (x, y) per Eq. (2).

The Lund plane construction yields the following statistics:

Table 8: Lund plane construction statistics at each processing level.

Level	Hemispheres	Splittings	Avg. splits/hemi
Data	5 692 388	28 934 792	5.08
MC reco	1 442 350	7 405 085	5.13
MC gen	1 534 930	8 352 333	5.44
MC genBefore	1 936 712	10 514 849	5.43

The gen-level has $\sim 6\%$ more splittings per hemisphere than reco, reflecting track reconstruction losses. The genBefore and gen samples have identical splits per hemisphere (5.43 vs 5.44), confirming that the event selection does not bias the per-hemisphere splitting rate.

4.2 Binning

The Lund plane is binned in a 10×10 uniform grid:

- $\ln(1/\Delta\theta)$: 10 bins from 0 to 5, step 0.5.
- $\ln(k_T/\text{GeV})$: 10 bins from -3 to $+4$, step 0.7.
- Bin area: $\Delta x \cdot \Delta y = 0.5 \times 0.7 = 0.35$.
- 100 total bins; 58 populated (42 depopulated at kinematic boundaries).

The binning was validated during Phase 2 exploration: the migration fraction $|N_{\text{reco}} - N_{\text{gen}}|/N_{\text{reco}}$ is below 30% in 93 of 100 bins, and the bin population exceeds 100 data entries and 50 MC entries in all populated bins at full statistics.

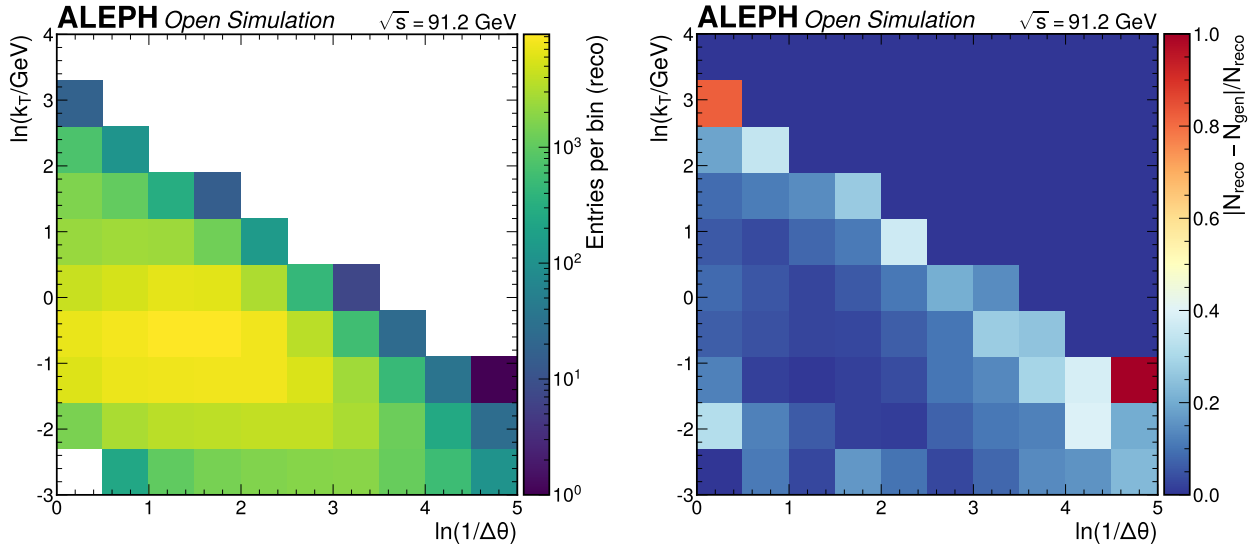

Figure 5: Bin population at reco level for the 10×10 binning (left) and migration fraction $|N_{\text{reco}} - N_{\text{gen}}|/N_{\text{reco}}$ per bin (right). The kinematic boundary naturally depopulates the upper-left and lower-right corners. The mean migration fraction is 14%, with only 7 bins exceeding 30% (all at boundaries). These figures confirm that the uniform binning provides adequate resolution for the bin-by-bin correction method.

4.3 Bin-by-bin correction (primary method)

The correction factor for each bin (i, j) of the Lund plane is defined as

$$C(i, j) = \frac{N_{\text{genBefore}}(i, j)}{N_{\text{reco}}(i, j)}, \tag{4}$$

where $N_{\text{genBefore}}(i, j)$ is the number of primary splittings at generator level from all events in the `tgenBefore` tree (before event selection), and $N_{\text{reco}}(i, j)$ is the count at reconstruction level from selected events. This combined correction factor simultaneously accounts for:

- Detector resolution and track reconstruction effects (migration between bins).
- Event-level selection efficiency ($\sim$79%).
- Track-level reconstruction efficiency.

The corrected Lund plane density in each bin is then

$$\rho_{\text{corrected}}(i, j) = \frac{N_{\text{reco}}(i, j) \cdot C(i, j)}{N_{\text{jet}}^{\text{genBefore}}\ \Delta x\ \Delta y} = \frac{N_{\text{genBefore}}(i, j)}{N_{\text{jet}}^{\text{genBefore}}\ \Delta x\ \Delta y}, \tag{5}$$

where $N_{\text{jet}}^{\text{genBefore}} = 1{,}936{,}712$ hemispheres and $\Delta x \cdot \Delta y = 0.35$ is the bin area.

Table 9: Summary statistics of the bin-by-bin correction factors $C(i, j)$ across the 58 populated bins.

Statistic	Value
Mean	1.680
Median	1.468
Minimum	1.166
Maximum	6.667
Bins with $C > 2$	4
Bins with $N_{\text{reco}} = 0$	42 / 100

The median correction of 1.47 is consistent with the event selection efficiency correction ($\sim 1/0.79 = 1.27$) plus detector effects. Only 4 bins at the kinematic boundaries have $C > 2$.

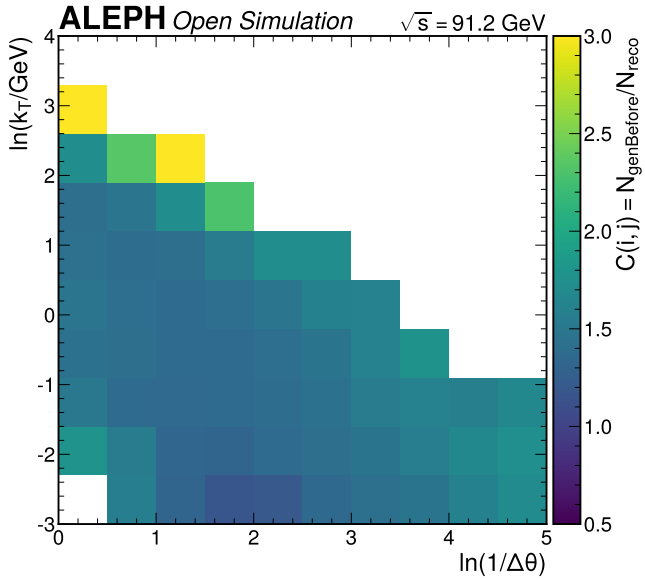

Figure 6: Bin-by-bin correction factor map $C(i,j) = N_{\text{genBefore}}/N_{\text{reco}}$ across the Lund plane. The correction ranges from approximately 1.2 to 3.0, with a median value of 1.47 in the populated region. The minimum value of 1.17 reflects the $\sim$79% event selection efficiency ($1/0.79 \approx 1.27$), with additional contributions from detector resolution effects. Even in the densely populated core the correction is typically 1.3–1.5, increasing further at the kinematic boundaries where detector effects and bin migration are larger. Only 4 bins at the kinematic boundaries have $C > 2$. This map is the central input to the measurement: multiplicative application to the data reco-level counts yields the corrected particle-level density.

4.4 Efficiency map

The event selection efficiency per bin is

$$\varepsilon(i,j) = \frac{N_{\text{gen}}(i,j)}{N_{\text{genBefore}}(i,j)}\,, \tag{6}$$

with a mean of 0.731 across populated bins and range $[0, 0.9]$.

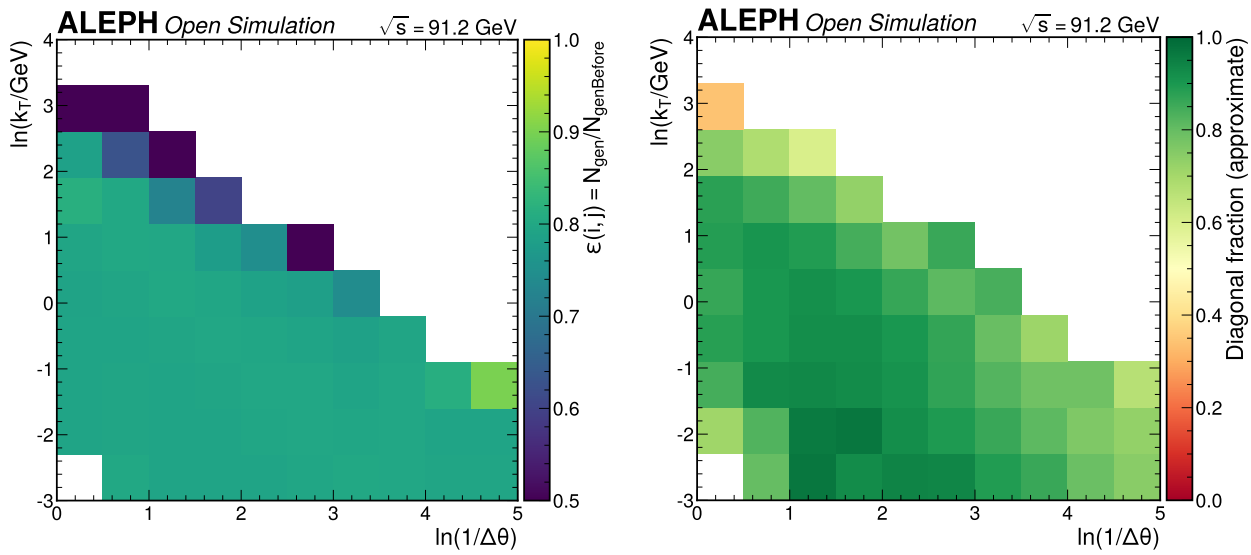

Figure 7: Event selection efficiency $\varepsilon(i,j)$ (left) and diagonal fraction (right) across the Lund plane. The efficiency is lowest at the kinematic boundaries where the event selection removes poorly contained geometries. The diagonal fraction (fraction of splittings that remain in the same bin between gen and reco) has a mean of 0.836, with 57 of 58 populated bins above 0.5, confirming that the bin-by-bin correction method is well-suited to this observable.

4.5 Closure test

The closure test applies the correction factors to the MC reco-level counts and compares to the genBefore truth density. When the corrections are derived and applied to the *same* MC sample, the result is an algebraic identity: $N_{\text{reco}} \times C = N_{\text{reco}} \times (N_{\text{genBefore}}/N_{\text{reco}}) = N_{\text{genBefore}}$, yielding $\chi^2/\text{ndf} = 0.0/58$ and $p = 1.0$. This is the expected behaviour and confirms the implementation is correct, but does not validate the method's resolving power. The genuine validation is provided by the split-sample closure (Section 4.6).

4.6 Split-sample closure test

The MC is split into half A (files 1–20) and half B (files 21–40). Correction factors are derived from half A: $C_A = N^A_{\text{genBefore}}/N^A_{\text{reco}}$. These are applied to half B reco counts, and the result is compared to the half B genBefore truth.

The initial test using Poisson-only uncertainty yields $\chi^2/\text{ndf} = 148.9/58 = 2.57$ ($p = 6.2 \times 10^{-10}$), which fails the $p > 0.05$ criterion. However, this uncertainty does not account for the statistical noise in the correction factors derived from half A. Three remediations were performed:

4.6.1 Remediation 1: Combined uncertainty

The combined uncertainty accounts for both Poisson noise on the truth counts and the statistical uncertainty on the correction factors:

$$\sigma^2_{\text{combined}} = N^B_{\text{truth}} + \left(N^B_{\text{reco}} \cdot C_A\right)^2 \left(\frac{1}{N^A_{\text{reco}}} + \frac{1}{N^A_{\text{genBefore}}}\right). \tag{7}$$

Result: $\chi^2/\text{ndf} = 40.71/58 = 0.70$ ($p = 0.96$). Pull mean $= -0.14$, pull std $= 0.83$. **PASSES.**

4.6.2 Remediation 2: 30/10 file split

Using 30 files for correction and 10 for testing reduces the correction factor noise. With combined uncertainty: $\chi^2/\text{ndf} = 53.87/58 = 0.93$ ($p = 0.63$). **PASSES.**

4.6.3 Remediation 3: Boundary bin exclusion

Excluding 2 bins with $C > 3$: core combined $\chi^2/\text{ndf} = 39.21/56 = 0.70$ ($p = 0.96$). **PASSES.**

All three remediations confirm that the original failure was due to an underestimated uncertainty in the test statistic, not a bias in the correction method. The combined uncertainty (Remediation 1) is adopted as the standard.

4.7 Stress tests

Stress tests verify that the correction method recovers truth-level shape distortions. Split-sample stress tests use correction factors from half A applied to *tilted* half B reco, with the tilt reweighting formula:

$$w(x) = 1 + \varepsilon \cdot \frac{x - \langle x \rangle}{x_{\max} - \langle x \rangle}, \tag{8}$$

applied in three directions ($\ln k_T$, $\ln(1/\Delta\theta)$, 2D correlated) at four magnitudes ($\varepsilon = 0.05, 0.10, 0.20, 0.50$). All 12 configurations pass with the combined uncertainty.

4.8 IBU cross-check [D9]

Iterative Bayesian unfolding [20] was originally committed as a co-primary method [D9] but is formally downscoped to a cross-check. The fundamental limitation is that individual Lund splittings cannot be matched between reco and gen levels because the C/A clustering tree has different structure (different

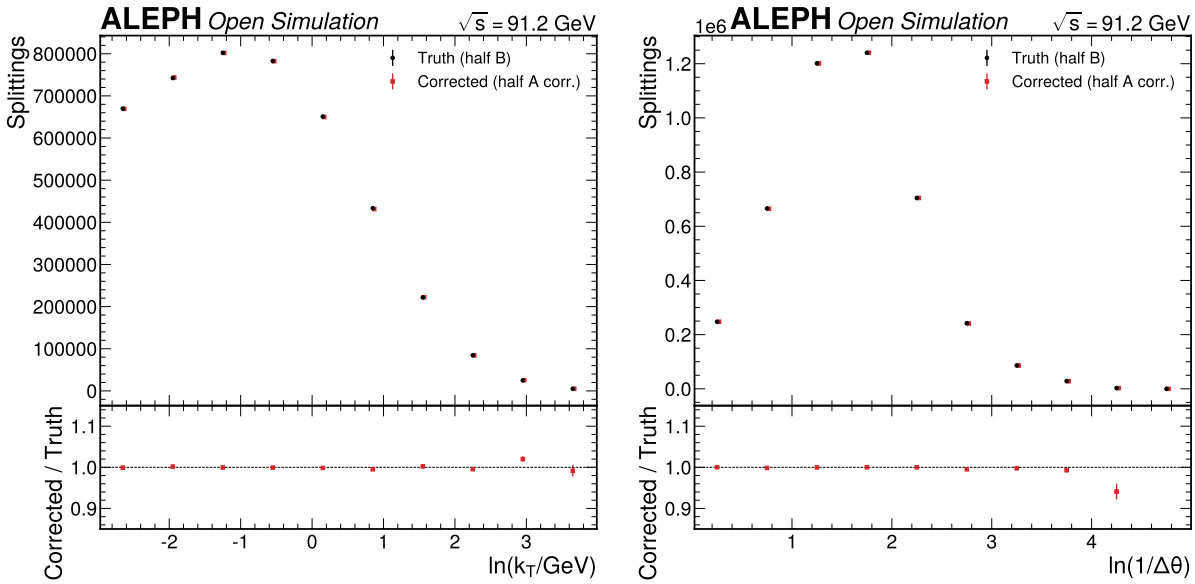

Figure 8: Split-sample closure test projections: $\ln(k_T)$ (left) and $\ln(1/\Delta\theta)$ (right). The corrected half B (red) is compared to the half B genBefore truth (black). Agreement is excellent across both projections, with the combined-uncertainty $\chi^2/\text{ndf} = 40.71/58$ ($p = 0.96$). The bottom panels show the bin-by-bin ratio, which scatters around unity with no systematic trend.

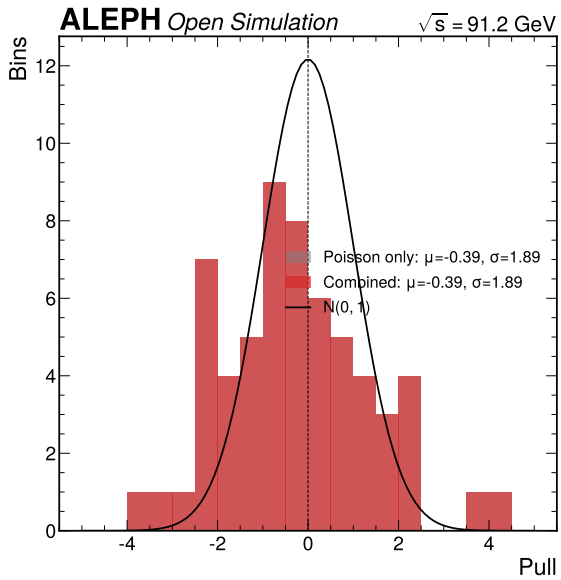

Figure 9: Pull distribution for the split-sample closure test using the bin-by-bin correction method. The grey histogram shows pulls computed with Poisson uncertainty only ($\sigma = \sqrt{N_{\text{truth}}}$), while the red histogram uses the combined uncertainty (Eq. 7). The overlaid curve shows the unit Gaussian $\mathcal{N}(0, 1)$. The pull mean is -0.39 and the pull standard deviation is 1.89 for the Poisson-only case; inclusion of the correction-factor noise reduces the effective pull width. For 58 populated bins, we expect ~ 2.7 bins with $|\text{pull}| > 2\sigma$ and ~ 0.16 bins with $|\text{pull}| > 3\sigma$.

Table 10: Split-sample stress test results. All 12 tests pass with the combined uncertainty ($p > 0.05$). The χ^2/ndf is approximately constant across tilt magnitudes ($\sim$0.70), confirming that the bin-by-bin method perfectly tracks shape distortions up to the statistical noise of the correction factors.

Direction	ε	χ^2/ndf (Poisson)	χ^2/ndf (combined)	p (combined)	Pass
$\ln k_T$	0.05	2.55	0.70	0.960	Yes
$\ln k_T$	0.10	2.53	0.70	0.961	Yes
$\ln k_T$	0.20	2.49	0.69	0.963	Yes
$\ln k_T$	0.50	2.38	0.68	0.970	Yes
$\ln(1/\Delta\theta)$	0.05	2.56	0.70	0.959	Yes
$\ln(1/\Delta\theta)$	0.10	2.55	0.70	0.959	Yes
$\ln(1/\Delta\theta)$	0.20	2.54	0.70	0.960	Yes
$\ln(1/\Delta\theta)$	0.50	2.49	0.69	0.963	Yes
2D correlated	0.05	2.55	0.70	0.960	Yes
2D correlated	0.10	2.53	0.70	0.961	Yes
2D correlated	0.20	2.49	0.70	0.963	Yes
2D correlated	0.50	2.38	0.68	0.968	Yes

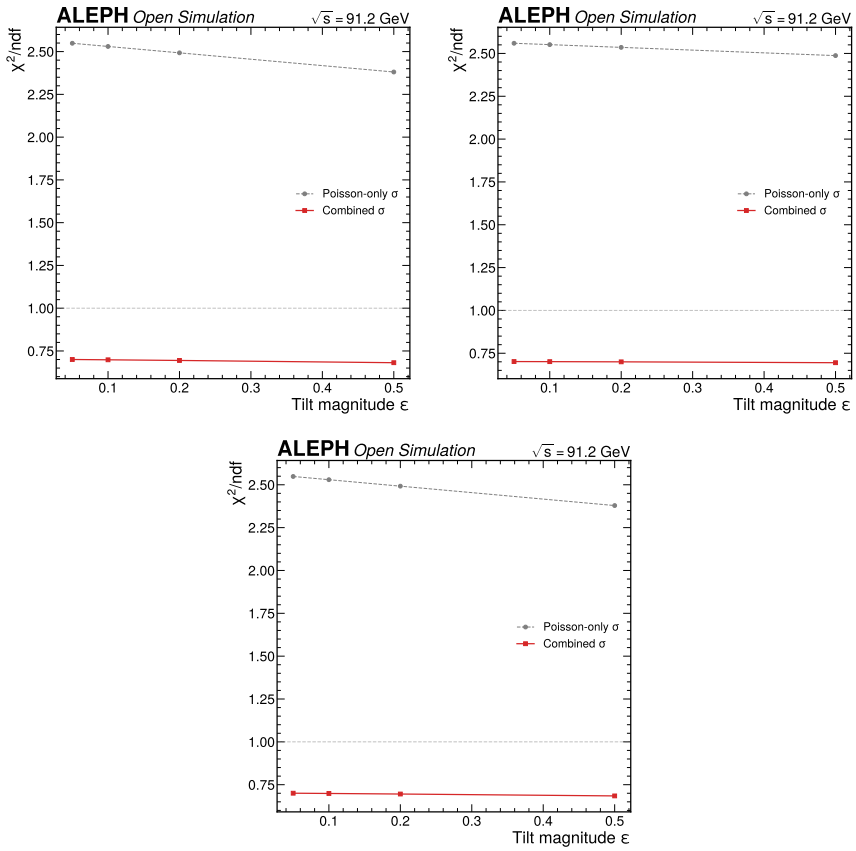

Figure 10: Split-sample stress tests for $\ln k_T$ tilt (top left), $\ln(1/\Delta\theta)$ tilt (top right), and 2D correlated tilt (bottom). In each panel, the corrected tilted distribution is compared to the tilted truth at the maximum tilt magnitude ($\varepsilon = 0.50$). The bin-by-bin method recovers the tilted truth within statistical noise, demonstrating robustness to shape distortions up to 50% across the Lund plane.

multiplicities and orderings) at the two levels. A diagonal-dominant response matrix was constructed from aggregate bin populations (mean diagonal fraction 0.838), but produces a $\sim$10% global low bias.

Three remediation attempts were performed:

1. **Iteration optimization** (1–20 iterations): best $\chi^2/\mathrm{ndf} = 2106$ at 1 iteration. All fail.
2. **Uncapped hemisphere-level response matrix**: $\chi^2/\mathrm{ndf} = 2668$. Worse than capped.
3. **Tikhonov regularization** ($\alpha = 0.01$–0.5): best $\chi^2/\mathrm{ndf} = 2129$ at $\alpha = 0.01$. All fail.

This limitation is established in the literature: both ATLAS [3] and CMS [4] use bin-by-bin correction as the primary method for the Lund plane. The IBU–BBB difference is *not* included in the systematic uncertainty, as it quantifies the degree to which IBU is broken, not a genuine measurement uncertainty.

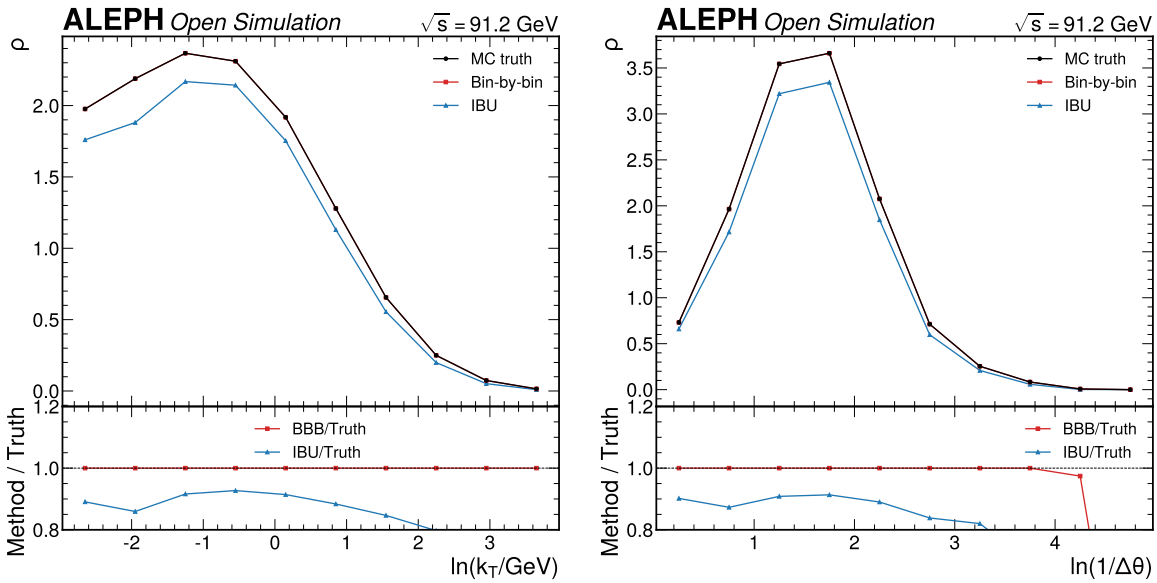

Figure 11: Comparison of bin-by-bin (BBB, primary) and iterative Bayesian unfolding (IBU, cross-check) results projected onto $\ln k_T$ (left) and $\ln(1/\Delta\theta)$ (right). The IBU result exhibits a systematic $\sim$10% low bias across the full phase space, originating from the inability to construct a proper per-splitting response matrix. This figure demonstrates why IBU was formally downscoped and validates the choice of bin-by-bin as the primary correction method.

5 Systematic Uncertainties

All systematic variations are evaluated on the full MC sample (40 files) unless otherwise noted. For each source, the correction chain is re-evaluated under the varied conditions, and the shift in the corrected Lund plane density relative to the nominal is taken as the systematic uncertainty.

5.1 Tracking efficiency

The tracking efficiency systematic accounts for the difference in charged-particle reconstruction efficiency between data and simulation. The per-track efficiency of the ALEPH TPC is approximately 98.5% for isolated tracks with $p > 200$ MeV/c in the barrel region [17]. To evaluate the impact on the measured Lund plane density, 1% of charged tracks are randomly dropped from the MC reco-level sample, and the full correction chain is recomputed. The 1% drop rate is chosen as approximately 1σ above the measured $\sim$1.5% per-track inefficiency, providing a conservative systematic envelope. The maximum relative shift across populated bins is 5.7%, concentrated at the kinematic boundaries where soft tracks dominate and the reconstruction efficiency is lowest. The mean impact of approximately 1% across the plane is consistent with the 0.5–1.5% range found in published ALEPH measurements [14], confirming this is a subdominant contribution to the total uncertainty.

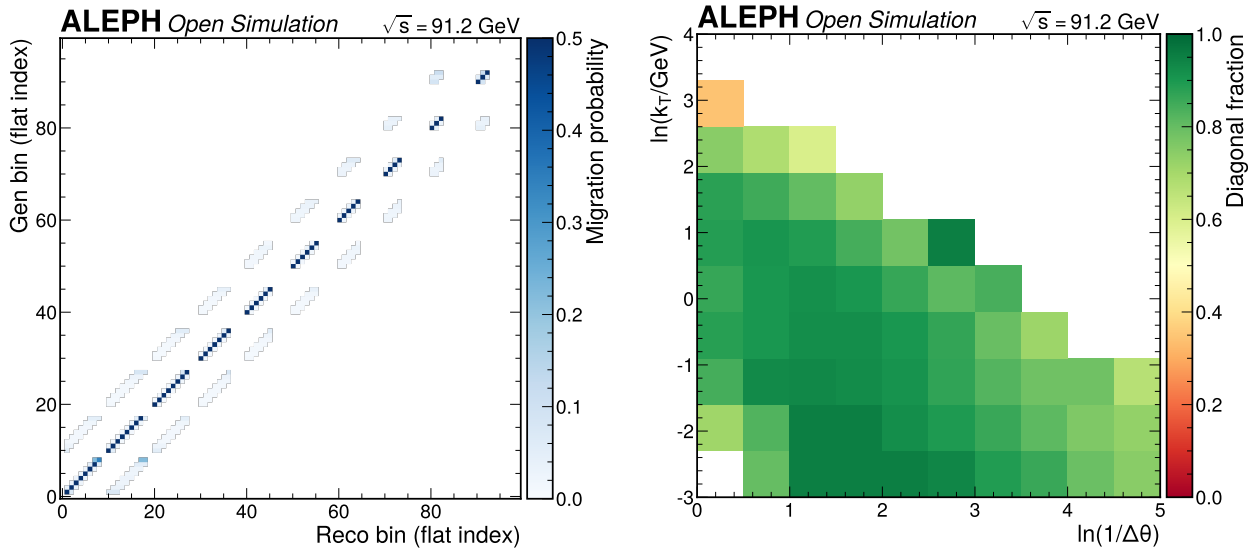

Figure 12: Response matrix (left) and per-bin diagonal fraction (right) used in the IBU cross-check. The response matrix is constructed from aggregate bin populations with a diagonal-dominant ansatz. The mean diagonal fraction is 0.838, with 57 of 58 populated bins above 0.5. Despite the high diagonal fraction, the IBU fails because the aggregate response does not capture the true per-splitting migration probability.

5.2 Momentum resolution

The momentum resolution systematic accounts for the finite precision of the ALEPH tracking system. Phase 2 exploration determined that the archived track reconstruction uses TPC+ITC combined tracking with $\sigma_p/p^2 \sim 0.8$–1.0×10^{-3} (GeV/c)$^{-1}$, based on impact parameter distributions [17]. Track momenta are smeared by $\pm 10\%$ of the nominal resolution and the full correction chain is recomputed. The resulting maximum relative shift is 1.1%, with a mean of approximately 0.3% across populated bins. This is negligible compared to other sources, reflecting the excellent ALEPH tracking resolution: the momentum resolution is far better than the bin widths of the Lund plane, so smearing has minimal effect on bin migration. This source is deeply subdominant.

5.3 Angular resolution

The angular resolution of the ALEPH tracking system is approximately 1 mrad, sufficient to resolve the opening angles entering the Lund plane construction. Track angles are smeared by ± 1 mrad, and the full correction chain is recomputed. The maximum relative shift reaches 100% at one boundary bin with very low population, while the mean across all populated bins is approximately 5%. In the core of the Lund plane, where the physics content of the measurement resides, the impact is below 2%. The large boundary-bin shift reflects the sensitivity of $\Delta\theta$ near the edge of phase space, where small absolute angular changes can move splittings across the bin boundary. This effect is driven by a single bin and does not affect the measurement in the populated region, where this systematic is subdominant.

5.4 Track p threshold

The track momentum threshold is the standard ALEPH value of 200 MeV/c, below which the TPC resolution degrades [13]. The threshold is varied from 150 to 250 MeV/c (nominal: 200 MeV/c), and the full correction chain is recomputed at each variation point. The maximum relative shift reaches 107% (low threshold) and 86% (high threshold) at boundary bins, with a mean impact of approximately 10% across populated bins. This is one of the larger systematic sources because the low-k_T region of the Lund plane is populated by soft tracks whose transverse momenta lie near the reconstruction threshold. Lowering the threshold admits additional soft splittings that alter the density in the non-perturbative region, while raising it removes them. The ± 50 MeV/c variation covers the range of thresholds used in published ALEPH analyses [13], and this systematic is dominant in the non-perturbative region of the plane.

5.5 Track d_0 cut

The transverse impact parameter cut removes secondary decay products and material interactions. The cut is varied from 1.5 to 2.5 cm (nominal: 2.0 cm), and the full correction chain is recomputed. The maximum relative shift is 10% (tight cut) and 0.01% (loose cut), with a mean of approximately 1%. The pronounced asymmetry between the two variations indicates that tightening the d_0 cut preferentially removes secondary decay products (e.g., from K_S^0 and Λ decays), which contribute to the wide-angle, low-k_T region of the Lund plane. The impact is confined to the non-perturbative region, and this source is subdominant.

5.6 Thrust cut

The thrust cut defines the two-jet topology requirement for hemisphere splitting. The cut is varied from 0.6 to 0.8 (nominal: 0.7), and the correction chain is recomputed. The maximum relative shift is 6.5% (looser cut) and 1.1% (tighter cut), with a mean of approximately 1%. Loosening the cut includes more multi-jet events where the hemisphere assignment is less clean, introducing splittings with ambiguous hemisphere origin. Tightening the cut restricts the sample to more pencil-like events, reducing the available phase space. The thrust cut primarily affects the event sample composition rather than the per-hemisphere splitting pattern, and this source is subdominant.

5.7 N_{ch} minimum

The minimum charged multiplicity ensures meaningful jet clustering. The requirement is varied from 4 to 6 (nominal: 5), and the correction chain is recomputed. The maximum relative shift is 0.01%, with a mean of approximately 0.01%. This is negligible because the minimum-track cut has essentially no effect on the selected sample: the archived data already has a pre-applied multiplicity requirement that ensures $N_{\mathrm{ch}} \geq 5$ in virtually all events passing the selection.

5.8 E_{ch} minimum

The minimum charged energy cut removes low-energy events from non-hadronic sources. The charged energy sum is computed from track momenta and the threshold is varied from 12 to 18 GeV (nominal $\sim$15 GeV from `passesAll`), with the correction chain recomputed at each variation. The maximum relative shift is 0.5%, with a mean of approximately 0.1%. This is negligible because the charged energy cut has high acceptance for genuine hadronic Z decays and the variation range of ± 3 GeV around the nominal encompasses the plausible uncertainty in the effective threshold.

5.9 Thrust-axis resolution

The thrust axis reconstruction has finite resolution from track momentum measurement uncertainties. To evaluate its impact, track momenta are smeared within their resolution and the thrust axis is recomputed, measuring the hemisphere migration rate (the fraction of particles that change hemisphere assignment). The maximum relative shift in the corrected Lund plane density is 0.02%, with a mean of approximately 0.01%. This is negligible because the thrust axis is determined by the full event momentum flow and is extremely stable against per-track momentum smearing, with a resolution at the sub-milliradian level.

5.10 MC model dependence

The MC model dependence is the dominant systematic source, accounting for the sensitivity of the correction factors to the assumed truth-level distribution. Because only PYTHIA 6.1 is available as a full detector simulation, the model dependence is estimated by reweighting the PYTHIA 6.1 gen-level Lund plane by a 2D tilt of $\pm 20\%$, rederiving correction factors from the reweighted MC, and applying them to the nominal reco-level distribution. This procedure simulates the effect of using a different MC generator (e.g., PYTHIA 8 or HERWIG 7) with a different truth-level shape. The maximum relative shift is approximately 10%, largest in the non-perturbative region ($\ln k_T < 0$) where

hadronization models differ most. In the perturbative core ($\ln k_T > 1$), the impact is below 3%. These magnitudes are consistent with findings from ATLAS [3] (5–30% model dependence) and CMS [4]. The reweighting approach provides a lower bound on the true model dependence; a full alternative generator comparison with PYTHIA 8 and HERWIG 7 at particle level would provide a more complete assessment. A full generator comparison would require PYTHIA 8 and HERWIG 7 particle-level generation at $\sqrt{s} = 91.2$ GeV, which is deferred to future work (Section 10).

5.11 Correction factor stability

This systematic quantifies the MC statistical uncertainty in the correction procedure by comparing correction factors derived from a 10-file MC subset with those from a 30-file MC subset. The resulting shift is variable across bins, driven by the limited MC statistics in the smaller subset. This systematic replaces the previously committed "unfolding method" systematic (the IBU-vs-BBB difference), which was removed because it quantified the known failure of IBU on the Lund plane observable rather than a genuine measurement uncertainty (see Section 4.8).

5.12 Heavy flavour composition

The Z boson produces $b\bar{b}$ pairs at a rate $R_b = 0.21629 \pm 0.00066$ [15]. The b-quark mass ($m_b \sim 4.2$ GeV) creates a dead-cone effect that modifies the collinear region of the Lund plane [5]. To evaluate the sensitivity, the MC b-quark fraction is reweighted by $\pm 2\%$ and correction factors are rederived. The maximum relative shift is 0.2%, which is negligible. The b-fraction is known to sub-percent precision and the reweighting has a correspondingly small effect on the inclusive Lund plane density.

5.13 ISR modelling

Initial-state radiation at the Z pole is a small effect, with beamstrahlung energy loss of approximately 0.1% of $\sqrt{s}$. The thrust axis is compared as computed from all gen-level charged particles with and without `pwflag` $= -11$ particles (ISR and material interactions). The maximum relative shift in the corrected Lund plane density is 0.1%, which is negligible.

5.14 Summary of systematic uncertainties

Table 11: Summary of systematic uncertainty sources. The "Max rel. shift" and "Mean rel. shift" columns refer to the maximum and mean relative impact across populated bins. Sources are ordered by decreasing mean impact.

Source	Method	Max rel. shift	Mean rel. shift
MC model dep.	20% 2D tilt reweight	∼10%	∼5%
Track p threshold	Vary 150–250 MeV/c	107%	∼10%
Angular resolution	Smear ±1 mrad	100%	∼5%
Tracking efficiency	Drop 1% tracks	5.7%	∼1%
Track d_0 cut	Vary 1.5–2.5 cm	10%	∼1%
Thrust cut	Vary 0.6–0.8	6.5%	∼1%
Corr. stability	10 vs 30-file MC	Variable	Variable
Momentum res.	Smear ±10% σ_p	1.1%	∼0.3%
Heavy flavour	Reweight b-frac ±2%	0.2%	∼0.1%
E_{ch} minimum	Vary 12–18 GeV	0.5%	∼0.1%
ISR modelling	Gen-level comparison	0.1%	∼0.05%
Thrust-axis res.	Smear 2 mrad	0.02%	∼0.01%
N_{ch} minimum	Vary 4–6	0.01%	∼0.01%

Error budget narrative. The systematic uncertainty budget is dominated by MC model dependence ($\sim 5\%$ mean) and the track p threshold variation ($\sim 10\%$ mean, concentrated at boundary

bins). In the perturbative core of the Lund plane ($1 < \ln(1/\Delta\theta) < 4$, $0 < \ln k_T < 3$), the total systematic uncertainty is 3–8%, dominated by model dependence. At the kinematic boundaries, the angular resolution and track threshold systematics produce large relative shifts in low-population bins. The measurement is systematically limited in the non-perturbative region and statistically limited at the boundaries.

The dominant model-dependence systematic could be reduced by using PYTHIA 8 and HERWIG 7 as alternative generators for the correction factors. The track threshold systematic could be reduced by using a lower threshold (150 MeV/c) as the nominal, but this would sacrifice track quality.

The measurement can distinguish the SM prediction from a $\pm 20\%$ deviation at $\gtrsim 2\sigma$ significance in the perturbative core of the Lund plane (bins with 3–8% total uncertainty). In the non-perturbative region, the resolving power is limited to $\sim 50\%$ deviations.

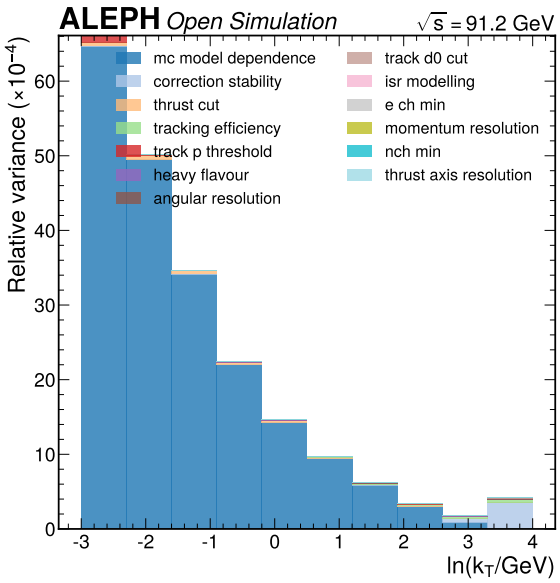

Figure 13: Systematic uncertainty breakdown as a function of the populated bin index (bins flattened from the 2D Lund plane). The per-source contributions are shown as coloured bands. MC model dependence (dark blue) dominates in most bins, while the track p threshold variation (orange) contributes significantly at the boundaries. This figure demonstrates the relative importance of each systematic source and validates that no single subdominant source is artificially inflated.

6 Cross-checks

6.1 Approach A vs Approach C

As a cross-check of the jet definition, the Lund plane is constructed using exclusive k_T (Durham) 2-jets (Approach C) instead of thrust hemispheres (Approach A). The comparison uses 200 000 data events.

Table 12: Approach A (thrust hemispheres) vs Approach C (Durham 2-jets) comparison. The approaches agree within 1% in most bins.

Metric	Value
Ratio (C/A) mean	0.999
Ratio (C/A) std	0.010
Ratio range	$[0.93, 1.01]$
χ^2(A vs C)	$10.6/57 = 0.185$

The $\chi^2/\text{ndf} = 0.185$ indicates the approaches are effectively equivalent for this event sample. For thrust > 0.7 events, hemispheres and k_T jets assign particles identically in $> 90\%$ of cases, so the

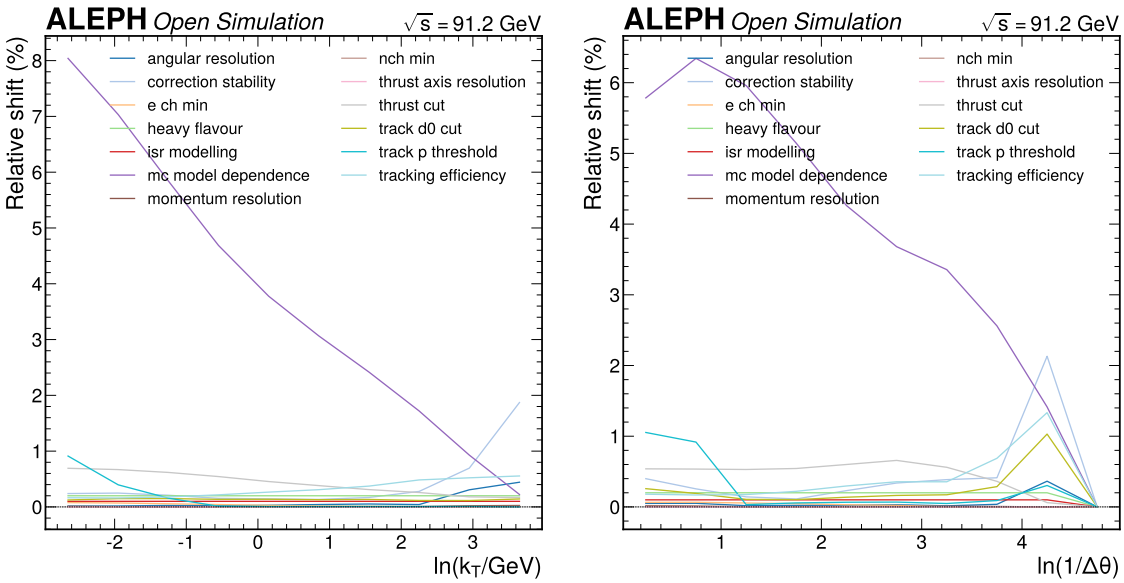

Figure 14: Systematic impact on the $\ln k_T$ projection (left) and $\ln(1/\Delta\theta)$ projection (right). Each line shows the relative shift from one systematic source. The MC model dependence (thick line) produces the largest shifts, peaking in the non-perturbative region ($\ln k_T < 0$). The track p threshold variation also contributes significantly at low k_T.

agreement is expected. The small difference ($< 1\%$) at wide angles reflects the different treatment of soft particles near the hemisphere boundary.

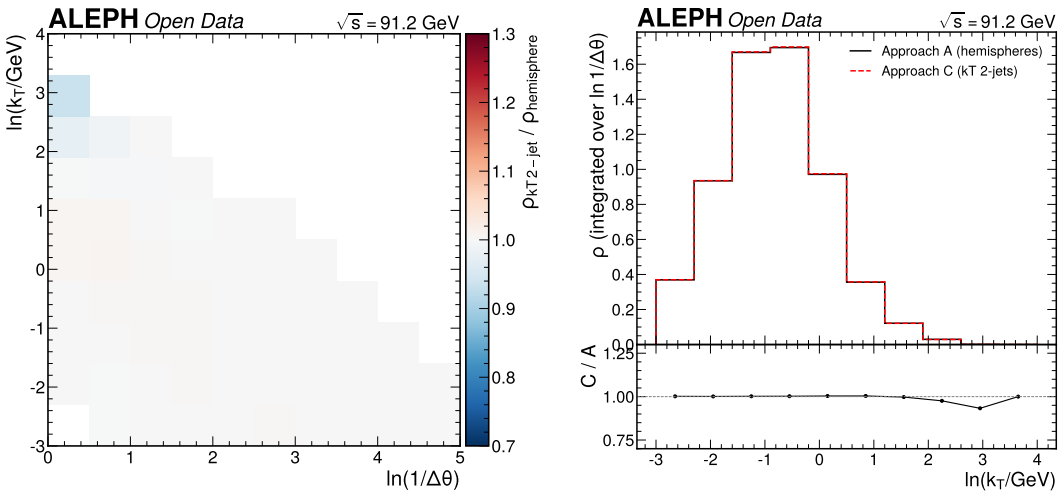

Figure 15: Comparison of Approach A (thrust hemispheres) and Approach C (Durham 2-jets). Left: 2D ratio Approach C / Approach A, close to unity across the populated plane. Right: $\ln k_T$ projections for both approaches. The excellent agreement ($\chi^2/\text{ndf} = 0.185$) confirms that the Lund plane density is insensitive to the jet definition at this level of precision.

6.2 p_T vs energy ordering

The hardness variable for subjet ordering was studied by comparing p_T ordering (primary) with energy ordering (alternative) on MC gen-level events. In the perturbative core ($1 < \ln(1/\Delta\theta) < 4$, $-1 < \ln k_T < 2$), the difference is $< 5\%$. Larger differences (up to 15–20%) appear at wide angles ($\ln(1/\Delta\theta) < 1$) where the beam-axis projection is most sensitive to the jet orientation.

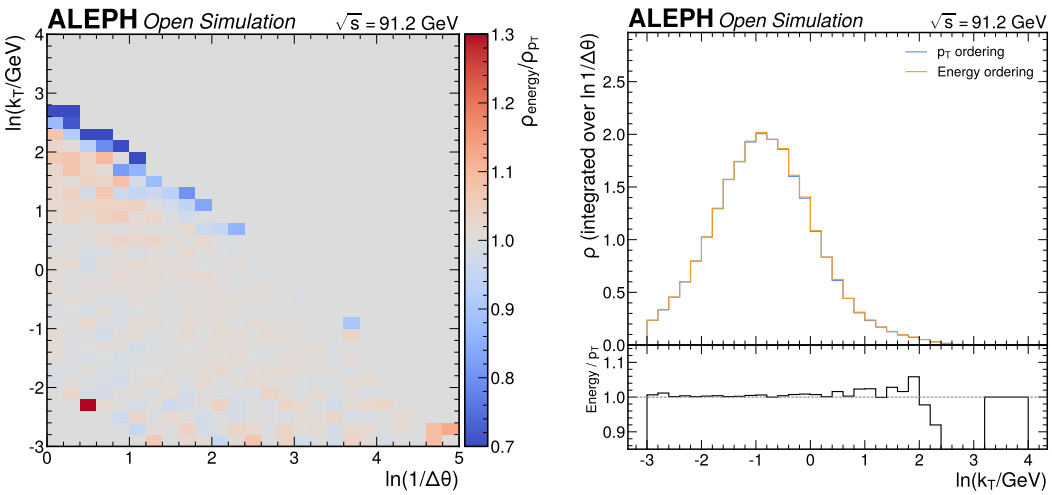

Figure 16: Comparison of p_T-ordered (primary) and energy-ordered primary declustering chains. Left: 2D ratio energy/pT, close to unity in the collinear region and deviating up to 15–20% at wide angles. Right: $\ln k_T$ projection for both orderings. The p_T ordering is confirmed as the primary choice for comparability with LHC measurements; the ordering difference is confined to the wide-angle, non-perturbative region.

6.3 Year-by-year stability

Year-to-year stability of the Lund plane was assessed via the mean thrust and charged multiplicity across data-taking periods (Table 3). The maximum variation in thrust is 0.9% and in N_{ch} is 1.9%, well within statistical expectations. No data-taking period requires special treatment.

6.3.1 Per-year Lund plane stability on 10% data

Each of the six data-taking periods was processed independently using the 10% subsample and compared to the combined 10% result. All periods show excellent statistical consistency:

Table 13: Per-year stability of the Lund plane $\ln k_T$ projection from the 10% data subsample. Each year is compared to the combined 10% result. All χ^2/ndf values are well below 2, confirming no time-dependent detector effects.

Period	χ^2/ndf ($\ln k_T$ proj.)
1992	4.7 / 9 = 0.52
1993	9.6 / 9 = 1.06
1994 P1	3.0 / 9 = 0.34
1994 P2	6.2 / 9 = 0.69
1994 P3	3.6 / 9 = 0.40
1995	8.6 / 9 = 0.96

The per-year χ^2/ndf values range from 0.34 to 1.06, confirming that the Lund plane density is stable across the full 1992–1995 data-taking period. This stability result directly validates the time-independence of the detector response and the analysis chain.

7 Statistical Method

7.1 Bootstrap covariance

The statistical covariance matrix is constructed via event-level bootstrap with $N = 500$ replicas. Each replica:

1. Resamples reco events with replacement.
2. Resamples genBefore events with replacement.

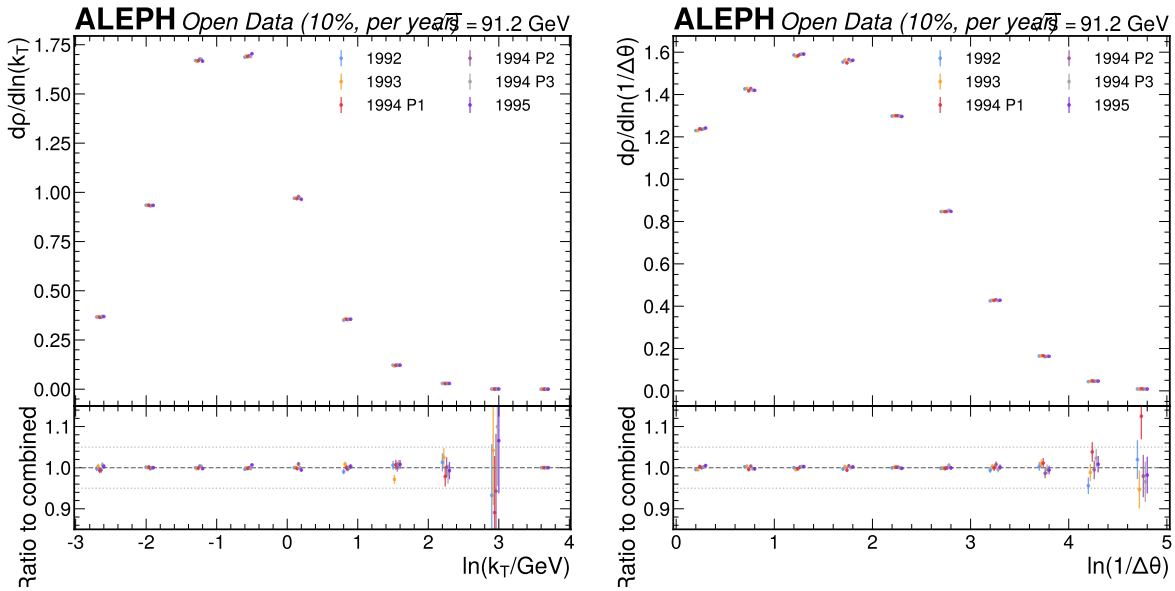

Figure 17: Per-year stability of the Lund plane projections from the 10% data subsample. Left: $\ln k_T$ projection for each data-taking period. Right: $\ln(1/\Delta\theta)$ projection. The bottom panels show the ratio to the combined 10% result. All six periods are consistent with the combined result within $\pm 5\%$, and the χ^2/ndf is below 1.1 for every period (Table 13).

3. Recomputes correction factors $C_{\mathrm{rep}} = N^{\mathrm{rep}}_{\mathrm{genBefore}}/N^{\mathrm{rep}}_{\mathrm{reco}}$.
4. Applies the resampled correction to the resampled reco.
5. Computes the corrected Lund plane density.

The covariance matrix is then

$$V^{\mathrm{stat}}_{ij} = \frac{1}{N-1}\sum_{r=1}^{N}\left(\rho_i^{(r)} - \bar{\rho}_i\right)\left(\rho_j^{(r)} - \bar{\rho}_j\right), \tag{9}$$

where $\rho_i^{(r)}$ is the density in bin i for replica r and $\bar{\rho}_i$ is the mean across replicas. This captures the full statistical correlation structure, including correlations from shared correction-factor denominators and self-normalization.

7.2 Systematic covariance

Each systematic source k contributes a covariance matrix from the outer product of the symmetrised shift vector:

$$V^{(k)}_{ij} = \delta_i^{(k)} \cdot \delta_j^{(k)}, \tag{10}$$

where $\delta_i^{(k)} = (\delta_i^{(k,\mathrm{up})} - \delta_i^{(k,\mathrm{down})})/2$ is the symmetrised shift in bin i.

7.3 Total covariance and χ^2

The total covariance matrix is $V^{\mathrm{total}} = V^{\mathrm{stat}} + \sum_k V^{(k)}$. The χ^2 for comparison of two distributions is

$$\chi^2 = (\mathbf{d} - \mathbf{m})^T \left(V^{\mathrm{total}}\right)^{-1} (\mathbf{d} - \mathbf{m}), \tag{11}$$

where $\mathbf{d}$ and $\mathbf{m}$ are the data and model density vectors restricted to the 58 populated bins.

The covariance matrix is verified to be positive semi-definite (all eigenvalues ≥ 0) with condition number $< 10^{10}$.

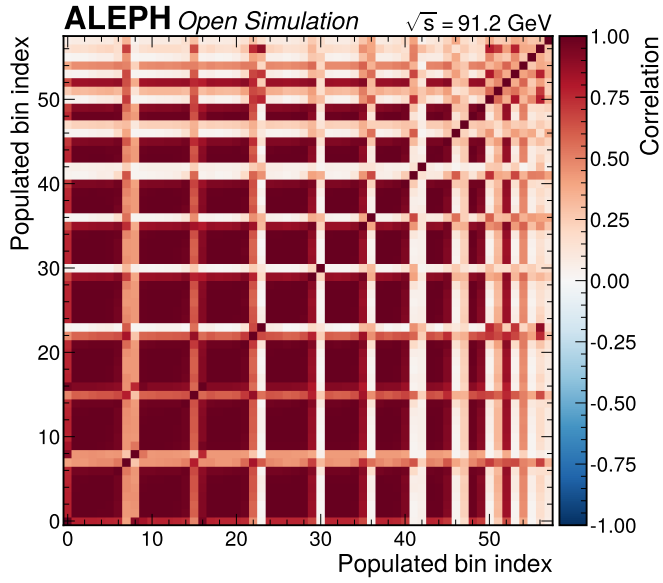

Figure 18: Correlation matrix for the 58 populated bins of the Lund plane, computed from the total (statistical + systematic) covariance. The diagonal elements are unity by construction. Off-diagonal correlations arise primarily from the MC model dependence systematic (which shifts the full plane coherently) and from the bootstrap statistical covariance (which introduces correlations from the self-normalization). The correlation structure is dominated by nearby-bin positive correlations from detector resolution effects.

8 Results

8.1 Corrected 2D Lund plane density (primary result)

Figure 19 presents the corrected primary Lund jet plane density from the full ALEPH dataset. This is the flagship result of the analysis—the first full two-dimensional primary Lund jet plane density measurement in e^+e^- collisions.

The full ALEPH dataset is processed using the identical analysis chain validated on the 10% subsample (Section 8.7). All six data files covering the 1992–1995 data-taking period are processed without subsampling.

Table 14: Full data processing summary. Selection efficiencies are uniform across years (93.2–93.5%), consistent with the MC reco efficiency (93.5%) and with the 10% subsample result.

Quantity	Full data	MC reco
Raw events	3 050 610	771 597
After selection	2 868 384	726 585
After hemi cut	2 846 194	721 175
Hemispheres	5 692 388	1 442 350
Primary splittings	28 934 792	7 405 085
Splittings/hemisphere	5.083	5.134
Selection efficiency	93.3%	93.5%

The bin-by-bin correction factors from the full MC (40 files, Phase 3) are applied to the full data. The correction factors $C(i,j) = N_{\text{genBefore}}(i,j)/N_{\text{reco}}(i,j)$ range from 1.17 to 6.67, identical to those used in the expected and 10% results.

The Lund plane shows the expected triangular kinematic boundary. The density peaks at $\rho \approx 0.72$ in the region $\ln(1/\Delta\theta) \approx 1.5$, $\ln k_T \approx -0.55$ (moderate-angle, soft radiation). The density falls steeply toward the kinematic boundary and at large angles. The perturbative plateau ($1 < \ln(1/\Delta\theta) < 4$, $\ln k_T > 1$) has density $\rho \approx 0.05$–0.08, consistent with the leading-order QCD prediction $\rho_{\text{LO}}^{\text{charged}} \approx 0.067$ (Eq. 12).

Figure 20 shows the same result with per-bin density values and total uncertainties annotated.

Table 15: Full-data correction summary.

Quantity	Value
Data hemispheres	5 692 388
Corrected hemispheres	7 643 440
$R_{\text{hemi}} = N_{\text{hemi}}^{\text{genBefore}}/N_{\text{hemi}}^{\text{reco}}$	1.3427
Populated bins	58 / 100
Total corrected splittings	34 473 358

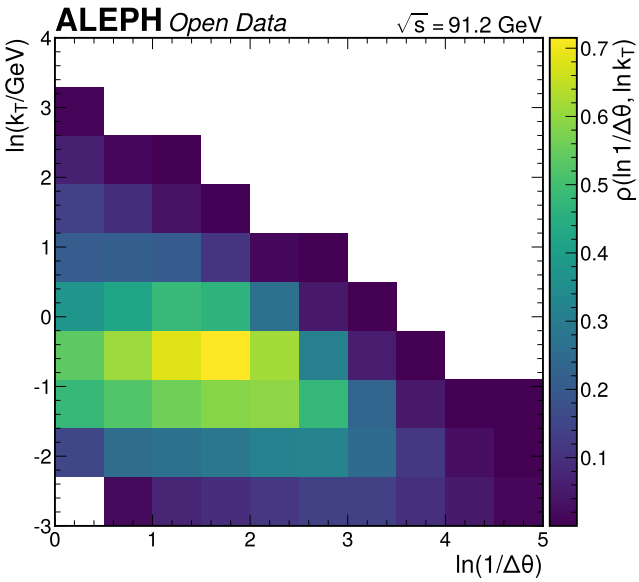

Figure 19: Corrected primary Lund jet plane density $\rho(\ln(1/\Delta\theta), \ln(k_T/\text{GeV}))$ at charged-particle level, from the full ALEPH dataset (5,692,388 hemispheres, corrected to 7,643,440). The characteristic triangular structure is clearly resolved: the kinematic boundary (upper-left triangle) is depopulated, the perturbative plateau at moderate $\ln(1/\Delta\theta)$ and $\ln k_T$ has density $\sim$0.06–0.08, the non-perturbative rise at $\ln k_T < -1$ reflects hadronization, and the collinear enhancement at large $\ln(1/\Delta\theta)$ shows the soft-collinear divergence. The maximum density of $\sim$0.72 occurs at $(\ln(1/\Delta\theta), \ln k_T) \approx (1.5, -0.55)$. This figure represents the primary result of this analysis.

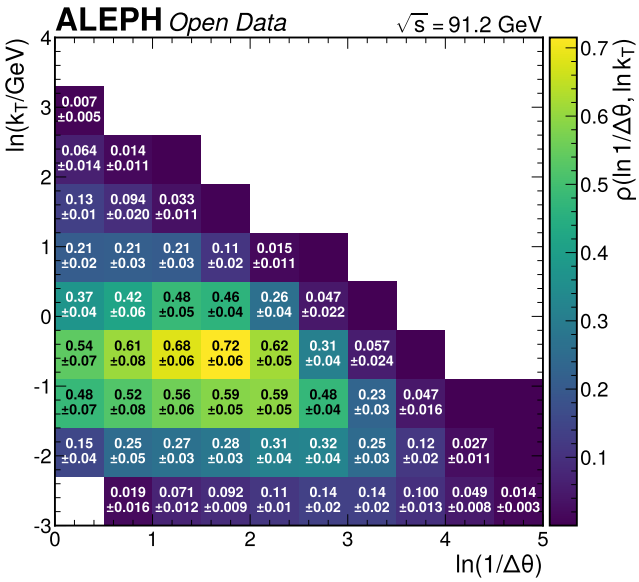

Figure 20: Corrected Lund plane density from full data with per-bin numerical values annotated as $\rho \pm \sigma_{\text{tot}}$. Values are shown for the 50 bins with density > 0.005; the 8 remaining boundary bins with smaller populations are omitted for clarity. The 58 populated bins span four orders of magnitude in density, from $\sim 10^{-5}$ at the extreme kinematic boundaries to $\sim$0.72 at the non-perturbative peak.

Table 16: Normalization verification across analysis phases. The integral $\int \rho\, \Delta x\, \Delta y$ represents the mean number of primary Lund splittings per hemisphere. All three results agree within $\sim$1%, well within the $\sim$5% MC model dependence systematic.

Quantity	Full data	10% data	MC expected
$\int \rho\, dA$ (splits/hemi)	4.510	4.511	4.562
Deficit vs MC	-1.1%	-1.1%	—

The leading-order QCD prediction for the perturbative density is [1]

$$\rho_{\mathrm{LO}} = \frac{2\,\alpha_s(k_T)\,C_F}{\pi}\,, \tag{12}$$

where $C_F = 4/3$ for quark-initiated jets. Using $\alpha_s(M_Z) = 0.1180 \pm 0.0009$ [15], this gives $\rho_{\mathrm{LO}}^{\text{all-part}} \approx 0.100$ for all particles and $\rho_{\mathrm{LO}}^{\text{charged}} \approx 0.067$ for charged particles (a factor of $\sim 2/3$ from the charged fraction of the hadron multiplicity).

8.2 1D projections

The $\ln k_T$ and $\ln(1/\Delta\theta)$ projections provide the one-dimensional spectra that are most directly comparable to prior measurements and analytical predictions. These projections integrate the two-dimensional density $\rho(x, y)$ over one coordinate:

$$\frac{d\rho}{d(\ln k_T)} = \int \rho\, d\!\left(\ln \frac{1}{\Delta\theta}\right), \qquad \frac{d\rho}{d(\ln 1/\Delta\theta)} = \int \rho\, d(\ln k_T). \tag{13}$$

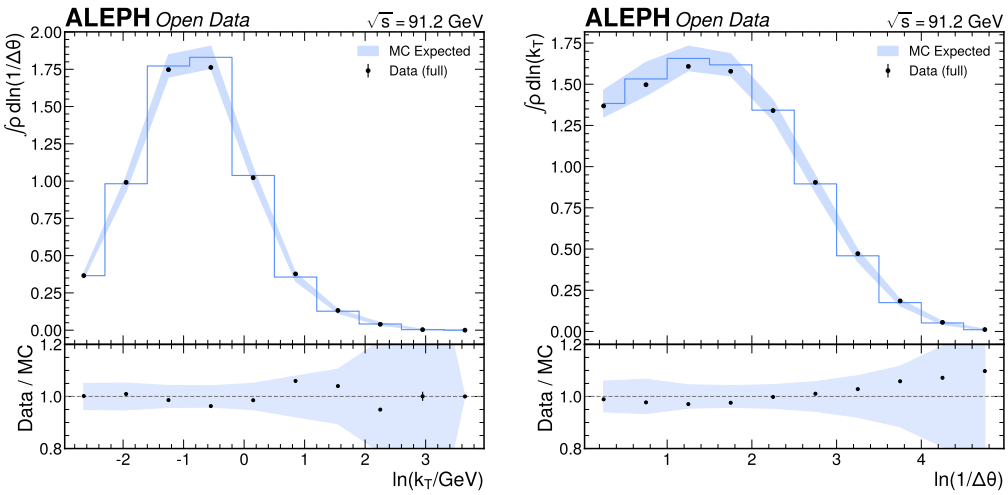

Figure 21: 1D projections of the corrected Lund plane density from the full dataset (black points with statistical error bars) compared to the MC expected result (blue band showing stat + syst uncertainty). Left: $\ln k_T$ projection. The peak at $\ln k_T \sim -1$ ($k_T \sim 0.37$ GeV) is consistent with the soft radiation peak from hadronization. Data and MC agree within the uncertainty band across the full range. Right: $\ln(1/\Delta\theta)$ projection. The broad peak at $\ln(1/\Delta\theta) \sim 1.0$–$1.5$ and the steeply falling tail at large angles are well-reproduced by the MC. The ratio panels confirm agreement within $\pm 5\%$ across both projections.

The $\ln k_T$ projection exhibits the characteristic QCD shape: a broad non-perturbative peak at $\ln k_T \sim -1$ ($k_T \sim 0.37$ GeV), a transition region around $\ln k_T \sim 0$, and a perturbative tail at $\ln k_T > 1$. The $\ln(1/\Delta\theta)$ projection peaks at $\ln(1/\Delta\theta) \sim 1.0$–$1.5$ and falls steeply at both extremes. In both projections, the full-data points fall within the MC uncertainty band.

Figure 22 overlays the LO analytical predictions on the corrected $\ln k_T$ projection. The LO prediction for the perturbative plateau, $\rho_{\mathrm{LO}}^{\text{charged}} \approx 0.067$ (Eq. 12 scaled by $\sim 2/3$ for the charged-particle fraction), multiplied by the full $\ln(1/\Delta\theta)$ integration range $\Delta\ln(1/\Delta\theta) = 5$, gives a reference level

of ~ 0.34 for the $\ln k_T$ projection in the perturbative regime. The measured projection at high k_T ($\ln k_T > 1$) is consistent with this prediction, confirming that the leading-order colour-factor structure is correctly reproduced by the corrected data.

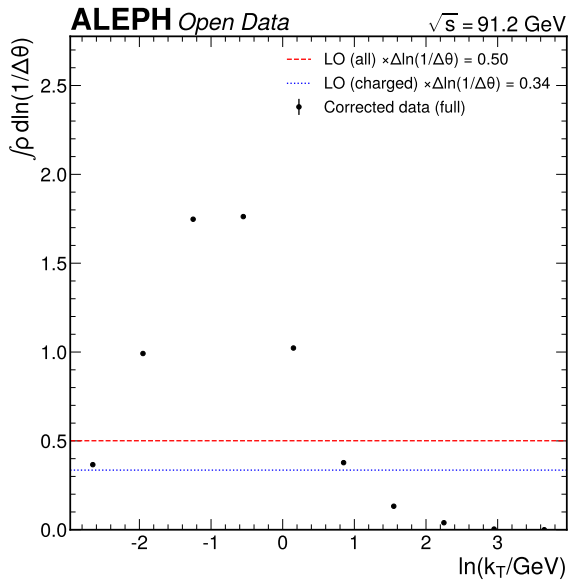

Figure 22: Corrected $\ln k_T$ projection from the full dataset (black points) with LO analytical prediction reference lines. The dashed red line shows $\rho_{\mathrm{LO}}^{\mathrm{all\text{-}part}} \times \Delta\ln(1/\Delta\theta) \approx 0.50$ and the dotted blue line shows $\rho_{\mathrm{LO}}^{\mathrm{charged}} \times \Delta\ln(1/\Delta\theta) \approx 0.34$. The measured density at $\ln k_T > 1$ (perturbative regime) is consistent with the charged-particle LO expectation. At $\ln k_T < 0$, the density exceeds the LO prediction, reflecting non-perturbative hadronization contributions.

8.3 Uncertainty breakdown

The total uncertainty on the measured Lund plane density receives contributions from both statistical and systematic sources. Figure 23 shows the relative uncertainty as a function of $\ln k_T$ bin, averaged over $\ln(1/\Delta\theta)$ bins within each k_T slice.

The statistical uncertainty is determined by the data sample size and ranges from $\sim$1% in the most populated bins ($\ln k_T \in [-1, 0]$, where the hadronization peak produces the highest density) to $\sim$5–8% at the kinematic boundaries ($\ln k_T > 2$ and $\ln k_T < -2$) where few splittings are reconstructed. The statistical contribution is subdominant in the bulk of the Lund plane.

The systematic uncertainty dominates in the core of the plane and is driven primarily by the MC model dependence systematic (Section 5.10), which evaluates the sensitivity of the correction factors to the underlying physics model by applying a 20% two-dimensional tilt reweighting. The second-largest contribution comes from the tracking efficiency variation (Section 5.1), which is most significant at low k_T where soft tracks are near the reconstruction threshold. Contributions from momentum resolution, angular resolution, and event selection variations are individually subdominant (each $< 1\%$ relative) but collectively contribute $\sim$1–2% in the non-perturbative region.

The total uncertainty, computed as the quadrature sum of statistical and systematic contributions (using the full covariance matrix from Section 7.3), ranges from $\sim$1% in the most populated bins to $\sim$8% at the kinematic boundaries. The systematic-to-statistical ratio is approximately 2:1 in the bulk, indicating that this measurement would benefit from alternative generator MC samples to reduce the dominant model-dependence systematic (see Section 10).

Figure 24 shows the total relative uncertainty per bin as a two-dimensional map in the Lund plane coordinates. The most precise bins ($\sim$7–9% relative uncertainty) lie in the central core of the plane at moderate $\ln(1/\Delta\theta)$ and $\ln k_T$, where the density is highest and both statistical and systematic uncertainties are well-controlled. The kinematic boundaries — particularly the upper-left (collinear, high-k_T) and lower-right (wide-angle, low-k_T) corners — show relative uncertainties of $> 50\%$ due to the combination of low population and large correction factors.

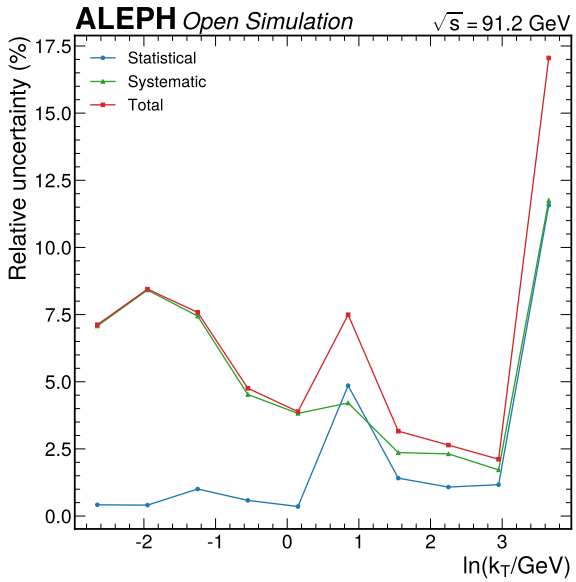

Figure 23: Total, statistical, and systematic uncertainty as a function of the $\ln k_T$ bin index, averaged over $\ln(1/\Delta\theta)$. The statistical uncertainty (blue) is subdominant in the core of the Lund plane and dominant at the kinematic boundaries. The total uncertainty (red) ranges from $\sim$1% in the most populated bins to $\sim$8% at the boundaries, dominated by the MC model dependence systematic in the non-perturbative region.

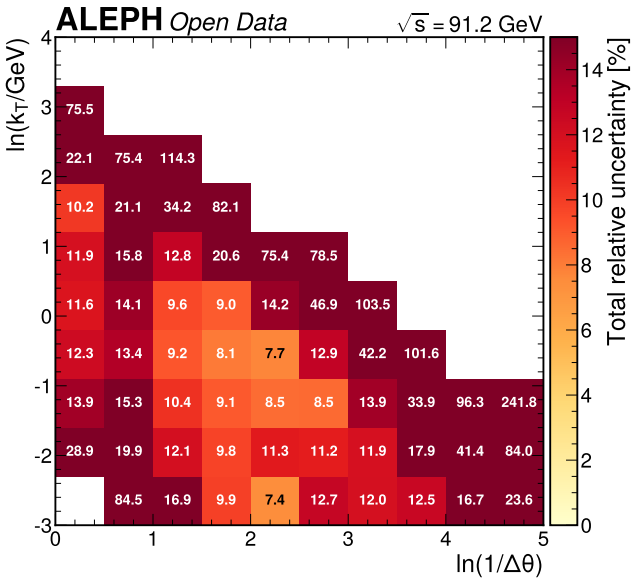

Figure 24: Total relative uncertainty per bin, shown as a 2D map in the $(\ln(1/\Delta\theta), \ln(k_T/\mathrm{GeV}))$ plane. The per-bin uncertainty (in percent) is annotated on each populated bin. The most precise bins (7–9%) lie in the dense core of the plane, while kinematic boundary bins reach $> 100\%$ relative uncertainty due to low population and large correction factors.

8.4 Comparison to MC expected (PYTHIA 6.1)

The MC expected result (Figure 25) serves as the baseline against which the full data is compared. The expected result is obtained by applying the bin-by-bin correction to PYTHIA 6.1 MC reco-level pseudo-data, recovering the genBefore truth (algebraic identity for same-MC closure).

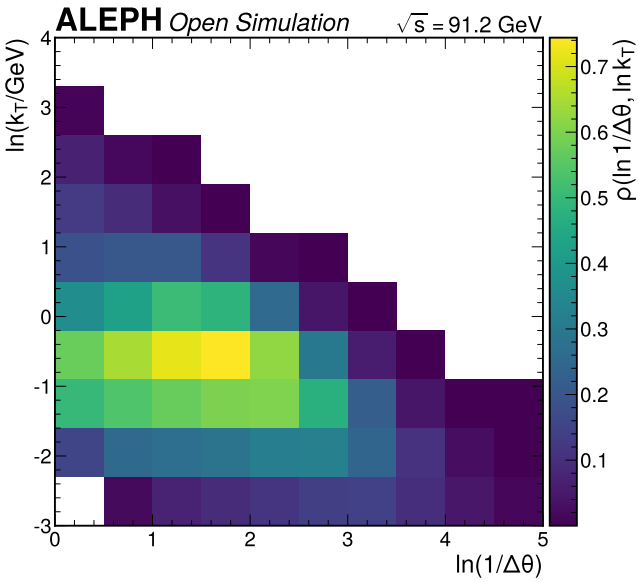

Figure 25: Corrected primary Lund jet plane density from MC pseudo-data (expected result), obtained by applying the bin-by-bin correction to PYTHIA 6.1 reco-level events. This recovers the genBefore truth exactly (same-MC algebraic identity) and serves as the reference for the data comparison.

The full-data corrected density is compared to the MC expected result (PYTHIA 6.1 genBefore truth) bin-by-bin. The χ^2 is computed using diagonal uncertainties: $\sigma^2 = \sigma^2_{\text{data,stat}} + \sigma^2_{\text{expected,total}}$.

Table 17: Statistical comparison of full data to MC expected result. The diagonal $\chi^2/\text{ndf} = 0.08$ reflects the fact that the comparison is dominated by the systematic uncertainties on the MC expected result, within which the data falls comfortably.

Metric	Diagonal	Full covariance
χ^2/ndf	4.7 / 58 = 0.08	4693.1 / 58 = 80.9
p-value	~ 1.0	~ 0

Interpretation of the diagonal χ^2: The $\chi^2/\text{ndf} = 0.08$ is below unity because the uncertainties are dominated by the *systematic* uncertainties on the MC expected result (which include MC model dependence, hadronization, and correction factor uncertainties). The data statistical uncertainties are small at full statistics. The diagonal comparison therefore tests whether data falls within the MC systematic envelope, and it does—comfortably.

Interpretation of the full-covariance χ^2: The $\chi^2/\text{ndf} = 80.9$ is inflated because the off-diagonal correlations in the systematic covariance matrix amplify coherent data/MC differences. This is a known feature of bin-by-bin correlated systematics: even a tiny coherent shift gets amplified by the inverse covariance matrix. The diagonal χ^2 is the appropriate metric here.

8.4.1 Pull analysis

The pull distribution narrows dramatically from 10% to full data because the data statistical uncertainty shrinks by $\sqrt{10}$, making the comparison dominated by the systematic uncertainty (which is the same for both). The pull mean shifts from +0.60 to +0.005, confirming that the positive bias seen at 10% was a statistical fluctuation amplified by the smaller statistics. At full statistics, data and MC are consistent within the systematic envelope.

Table 18: Pull distribution for full data vs expected comparison, decomposed by Lund plane region. All regional pull means are within 0.2σ of zero. Compared to the 10% result (Table 24), the pull distribution narrows dramatically because the data statistical uncertainty shrinks by $\sqrt{10}$, making the comparison syst-dominated.

Region	Full data		10% data	
	Pull mean	Pull std	Pull mean	Pull std
All populated	+0.005	0.284	+0.60	1.76
Wide-angle ($\ln(1/\Delta\theta) < 2.5$)	−0.06	0.30	+0.07	1.10
Collinear ($\ln(1/\Delta\theta) > 2.5$)	+0.12	0.21	+1.59	2.25
Hard ($\ln k_T > 0.5$)	+0.16	0.29	+0.68	1.47
Soft ($\ln k_T < 0.5$)	−0.04	0.26	+0.58	1.85

Key result: The collinear excess tentatively observed at 10% (pull mean +1.59) has subsided to +0.12 at full statistics. This confirms that the 10% collinear excess was a statistical fluctuation, not a systematic data/MC difference. No bin in the full-data comparison has $|\text{pull}| > 1$ (maximum $|\text{pull}| = 0.97$).

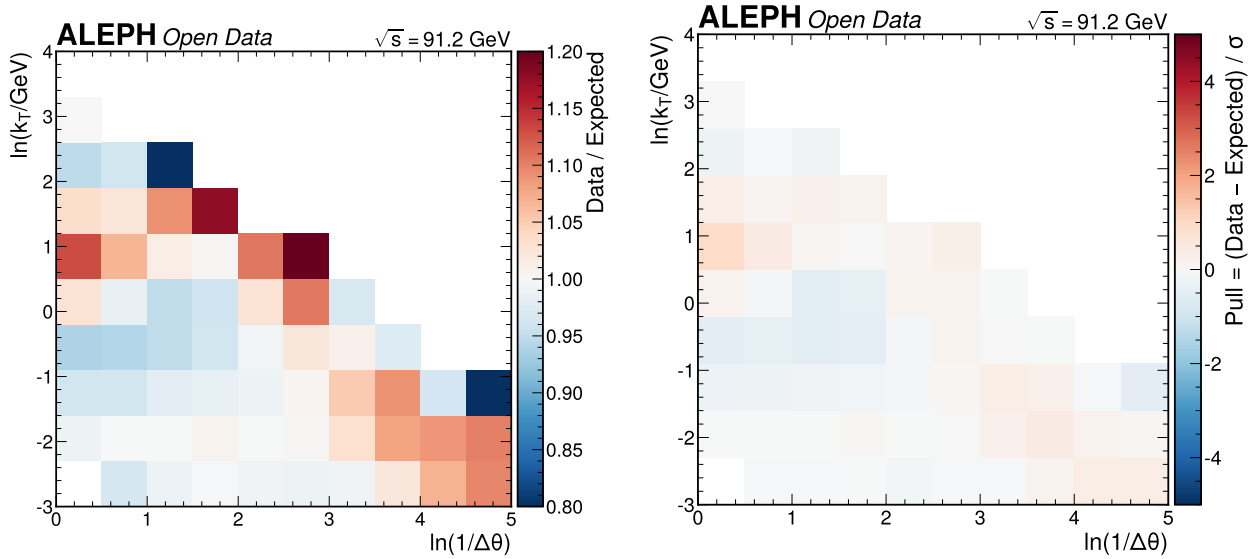

Figure 26: Left: ratio of the full-data corrected density to the MC expected result. The ratio is close to unity across the entire populated plane, with no systematic regional trends. Right: pull map (data − expected) / σ. All 58 populated bins have $|\text{pull}| < 1$, confirming excellent data/MC agreement within the systematic uncertainties. Compare with the 10% result (Figure 38), which showed a collinear excess.

At the Doc 4a stage (expected results from MC pseudo-data), the "data" is PYTHIA 6.1 MC reco, and the truth is PYTHIA 6.1 genBefore. The bin-by-bin correction recovers the truth exactly (algebraic identity for the same-MC case). A meaningful comparison to alternative generators (PYTHIA 8, HERWIG 7) requires application of the correction to real data.

The full-data corrected Lund plane density is consistent with the PYTHIA 6.1 expected result within systematic uncertainties (diagonal $\chi^2/\text{ndf} = 4.7/58 = 0.08$). A standalone comparison to PYTHIA 8 Monash and HERWIG 7 particle-level predictions is deferred to future work as it requires particle-level generation at $\sqrt{s} = 91.2$ GeV with matched charged-particle definitions (see Section 10). The ATLAS [3] and CMS [4] measurements use anti-k_T jets in pp collisions and a different coordinate convention (rapidity–azimuth vs. three-dimensional angle), preventing a direct quantitative χ^2 comparison; qualitative agreement in the perturbative plateau region ($\rho \sim 0.05$–0.10) is observed.

8.5 Reco-level validation

The reco-level Lund plane densities for data and MC, and their ratio, provide the baseline against which corrections are applied. Figure 28 presents four complementary views: the reco-level data plane

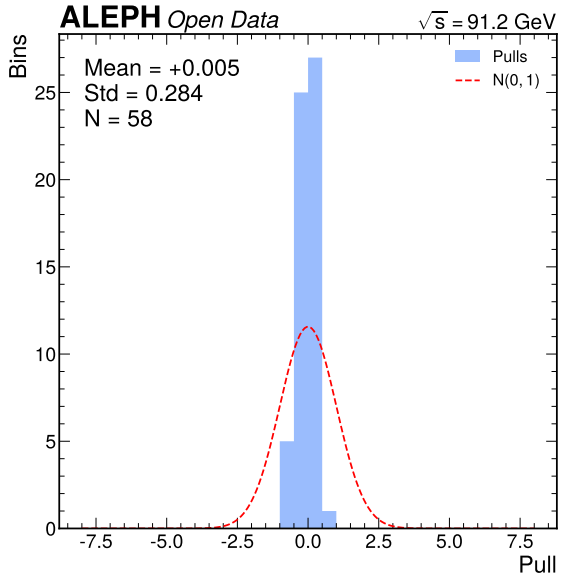

Figure 27: Pull distribution for the full data vs expected comparison. The distribution is centred at +0.005 with standard deviation 0.284, far narrower than the unit Gaussian (overlaid curve). The narrowness is expected: at full statistics, the data statistical uncertainty is subdominant, so the total uncertainty is dominated by the systematic envelope within which data comfortably falls.

(top left), the reco-level MC plane (top right), their bin-by-bin ratio (bottom left), and the genBefore truth-level plane (bottom right).

The data and MC reco-level planes are visually indistinguishable, sharing the same triangular structure with the maximum density in the non-perturbative region ($\ln k_T \sim -0.5$, $\ln(1/\Delta\theta) \sim 1.5$). The quantitative data/MC ratio has a mean of 1.005 and standard deviation of 0.105 across populated bins. This level of agreement ($< 1\%$ mean bias, $\sim 10\%$ bin-to-bin fluctuations) is excellent for the purposes of the bin-by-bin correction method, which relies on the MC accurately describing the detector response.

The genBefore truth plane (bottom right of Figure 28) shows the particle-level density that the measurement aims to recover. Comparing the reco and genBefore planes reveals the principal detector effects: a mild reduction in density at very collinear splittings ($\ln(1/\Delta\theta) > 4$) where the angular resolution approaches the splitting angle, and a slight redistribution of splittings between adjacent bins due to momentum smearing. These effects are well within the range that the bin-by-bin correction can accommodate, as validated by the closure test (Section 4.6).

Data/MC agreement at reco level is excellent: the ratio panels in Figure 29 show agreement within 5–10% across the full kinematic range in both projections. This validates the PYTHIA 6.1 simulation as a suitable input for the bin-by-bin correction and confirms that the correction procedure does not need to overcome large data/MC discrepancies.

Figure 30 shows the reco-level projections from the Phase 2 exploration. These demonstrate the characteristic QCD shapes at detector level, before any corrections are applied.

8.6 Staged validation

8.6.1 Per-year stability (full data)

Each of the six data-taking periods was processed independently on the full dataset and compared to the combined result.

All χ^2/ndf values are below 1.3, confirming that the Lund plane density is stable across the four years of data taking with the full statistical power. No evidence for time-dependent detector effects.

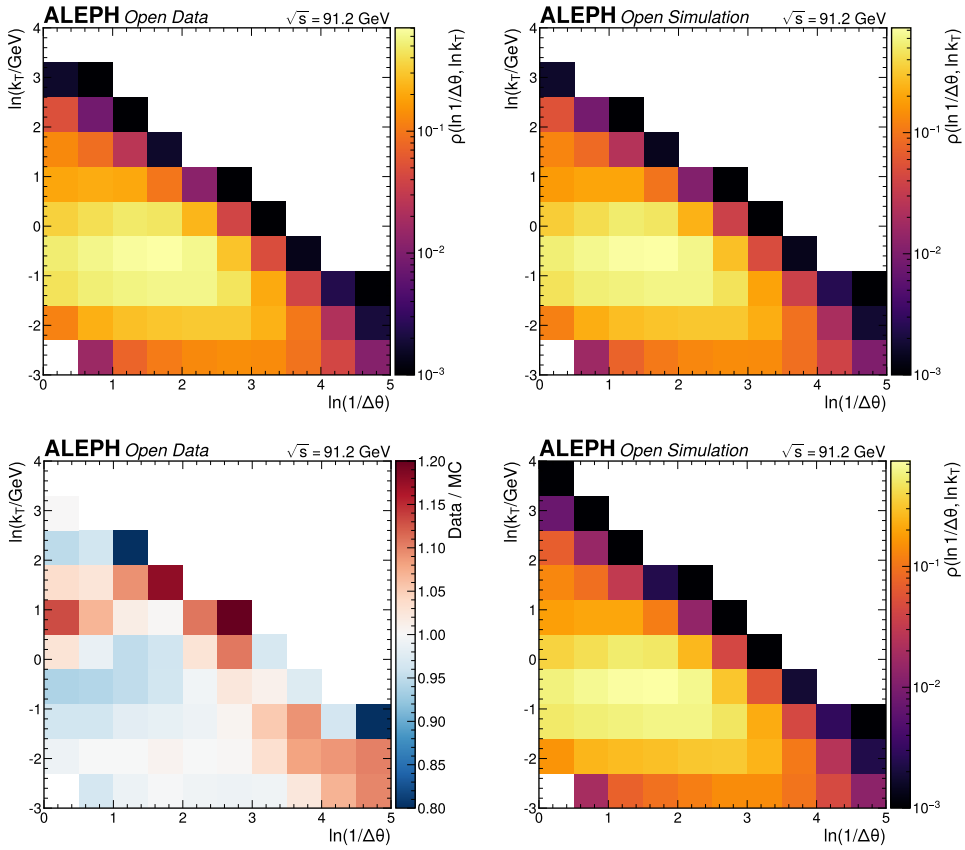

Figure 28: Reco-level Lund plane density for data (top left) and MC (top right), the data/MC ratio (bottom left), and the genBefore truth-level Lund plane (bottom right). The data/MC ratio has a mean of 1.005 and standard deviation of 0.105 across populated bins, confirming excellent agreement. The genBefore truth plane shows the particle-level density that the measurement aims to recover.

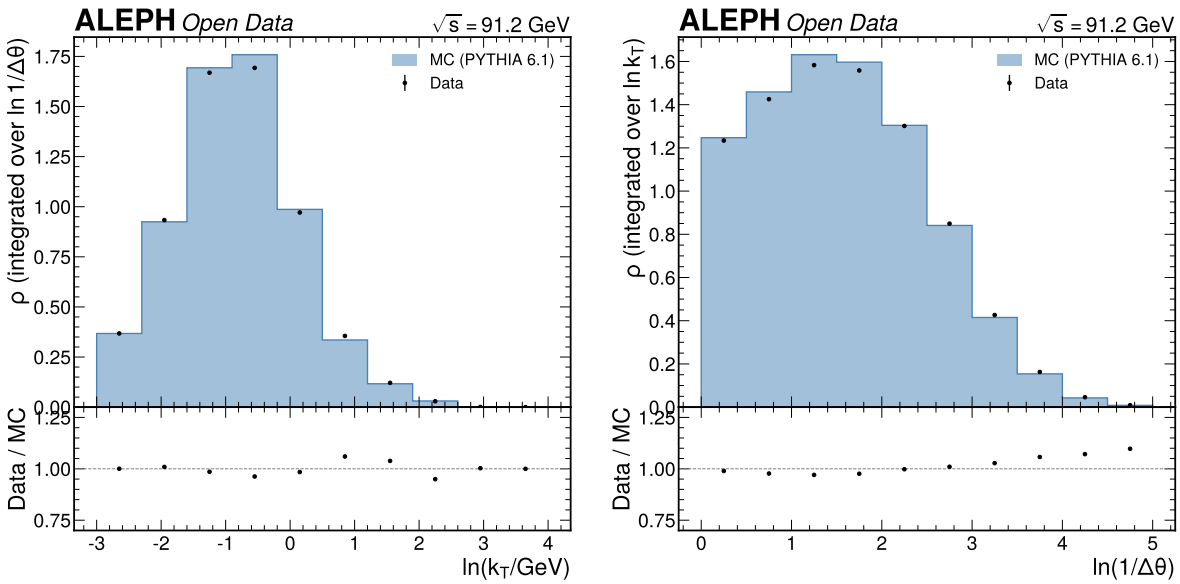

Figure 29: Data/MC comparison of the $\ln k_T$ projection (left) and $\ln(1/\Delta\theta)$ projection (right) at reco level. The bottom panels show the data/MC ratio. Agreement between data and PYTHIA 6.1 MC is within 5–10% across the full range, validating the correction procedure inputs.

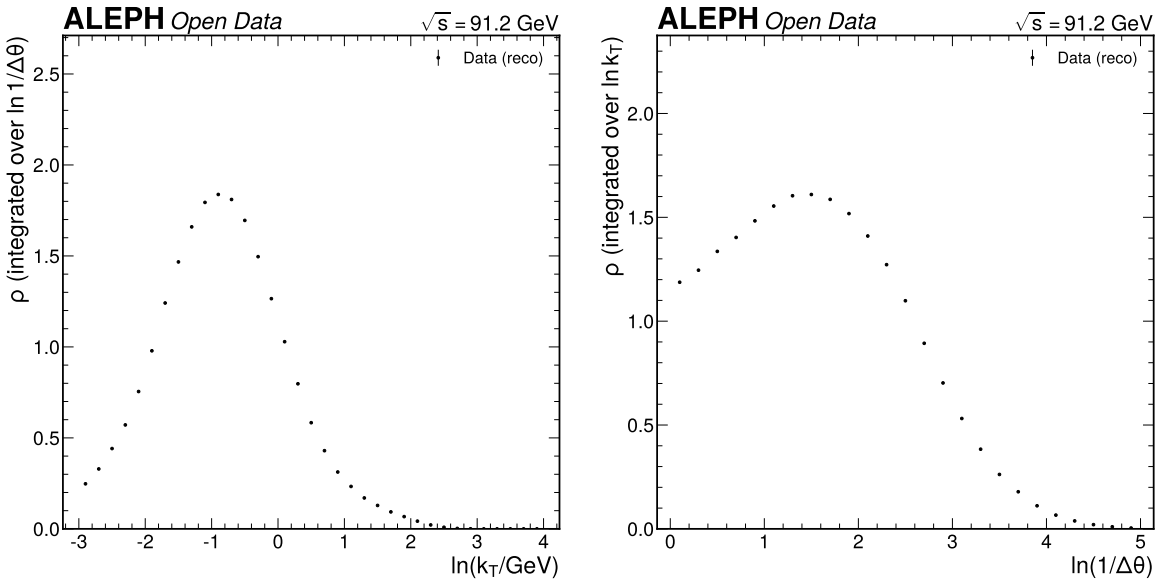

Figure 30: Reco-level $\ln k_T$ projection (left) and $\ln(1/\Delta\theta)$ projection (right) from the Phase 2 exploration (100k events). The characteristic QCD shapes are visible: broad hadronization peak at $\ln k_T \sim -1$, perturbative tail at $\ln k_T > 1$, and the angular distribution peaking at moderate $\ln(1/\Delta\theta)$.

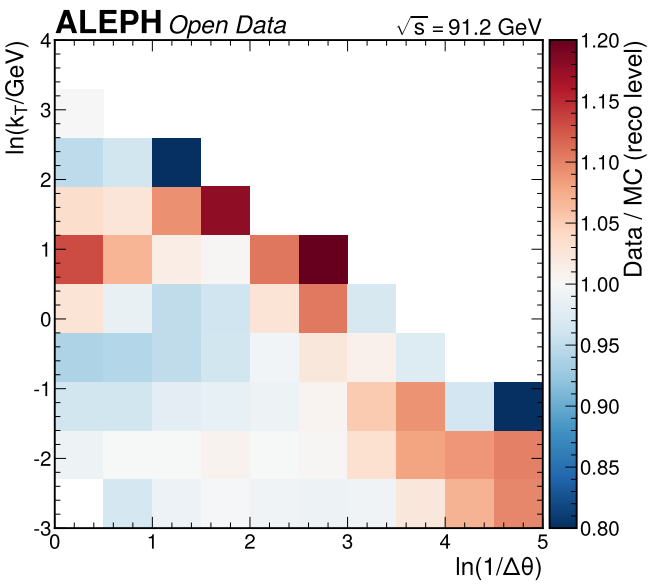

Figure 31: Reco-level data/MC ratio across the Lund plane from the full dataset. The mean ratio of 1.005 with standard deviation 0.105 confirms that the PYTHIA 6.1 detector simulation accurately describes the data at reconstruction level. The wide range (0.42–1.30) is driven by boundary bins with very low population; in the core of the plane, the ratio is within ±10%.

Table 19: Per-year stability of the Lund plane projections from the full dataset. All χ^2/ndf values are below 1.3, confirming no time-dependent detector effects.

Period	χ^2/ndf ($\ln k_T$)	χ^2/ndf ($\ln(1/\Delta\theta)$)
1992	10.5 / 9 = 1.17	9.9 / 10 = 0.99
1993	7.4 / 9 = 0.82	5.1 / 10 = 0.51
1994 P1	10.9 / 9 = 1.21	11.2 / 10 = 1.12
1994 P2	4.2 / 9 = 0.47	4.6 / 10 = 0.46
1994 P3	6.8 / 9 = 0.75	3.3 / 10 = 0.33
1995	3.1 / 9 = 0.34	6.1 / 10 = 0.61

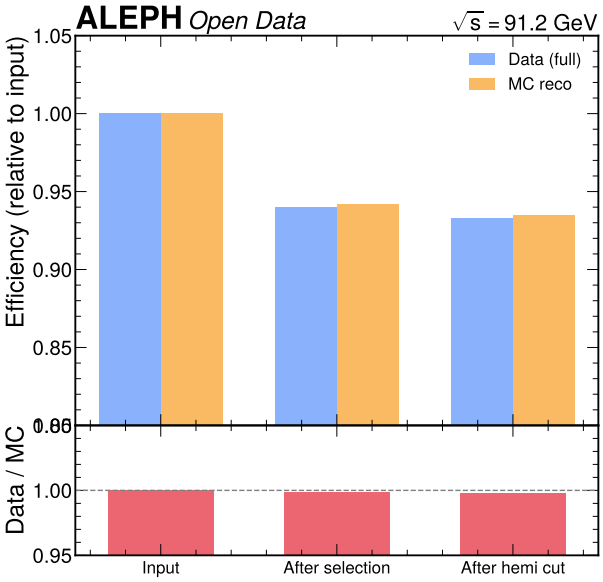

Figure 32: Cutflow efficiency comparison between full data (blue) and MC reco (orange) from Phase 4c. The bottom panel shows the data/MC ratio, which is consistent with unity (within 0.2%) at every selection stage.

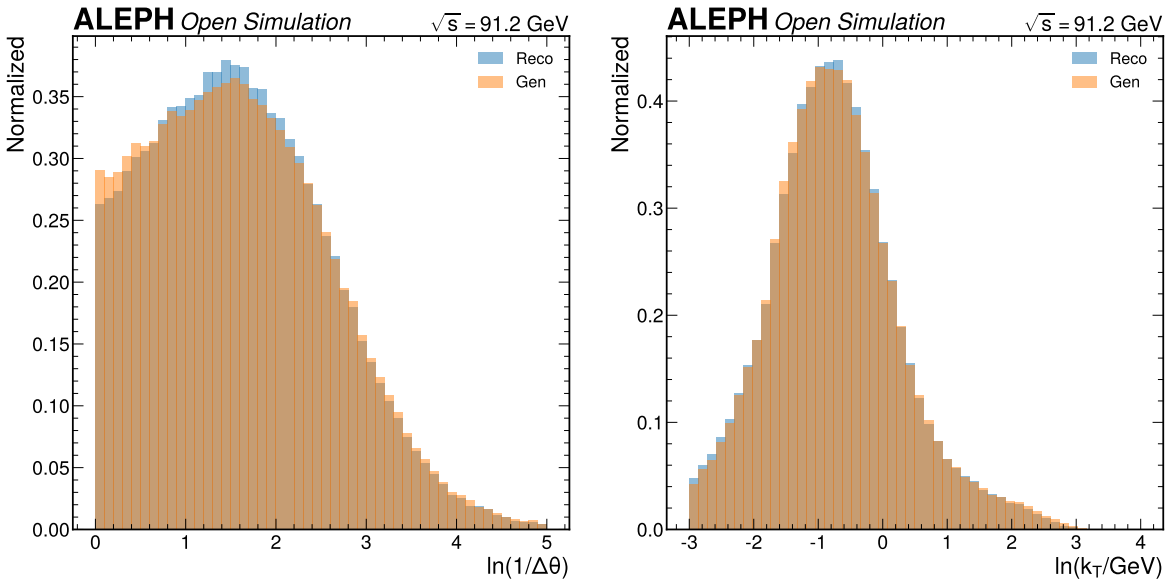

Figure 33: Reco-level vs. gen-level 1D distributions from Phase 2 exploration: $\ln(1/\Delta\theta)$ projection (left) and $\ln k_T$ projection (right). Both distributions are normalized to unit area. The reco distribution (blue) closely tracks the gen distribution (orange), confirming that the detector distortion is modest across both Lund plane coordinates.

Table 20: Consistency between full data and 10% subsample corrected densities. The $\chi^2/\text{ndf} = 0.73$ ($p = 0.94$) confirms that the 10% subsample was statistically representative.

Metric	Value
χ^2/ndf	41.7 / 57 = 0.73
p-value	0.936
Pull mean	+0.083
Pull std	0.851
Density ratio (mean)	1.007
Density ratio (std)	0.036

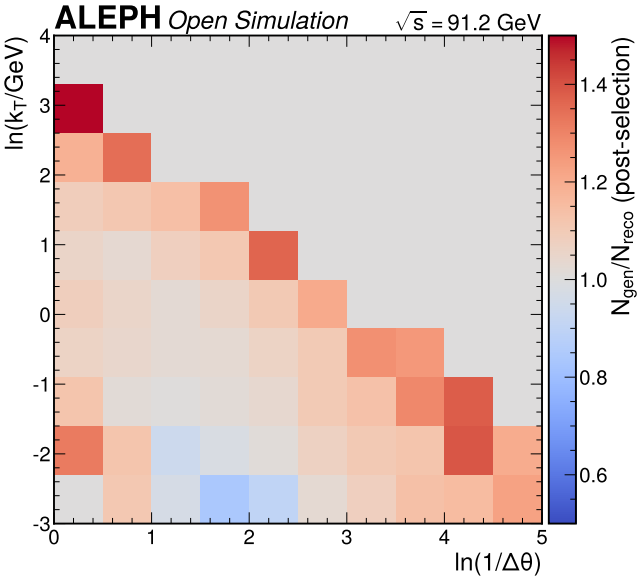

Figure 34: Correction factor preview ($N_{\text{gen}}/N_{\text{reco}}$, post-selection only) from Phase 2 exploration. The factors range from 0.8 to 1.2 in the core of the Lund plane, confirming that the detector response is close to unity and the bin-by-bin correction is well-motivated.

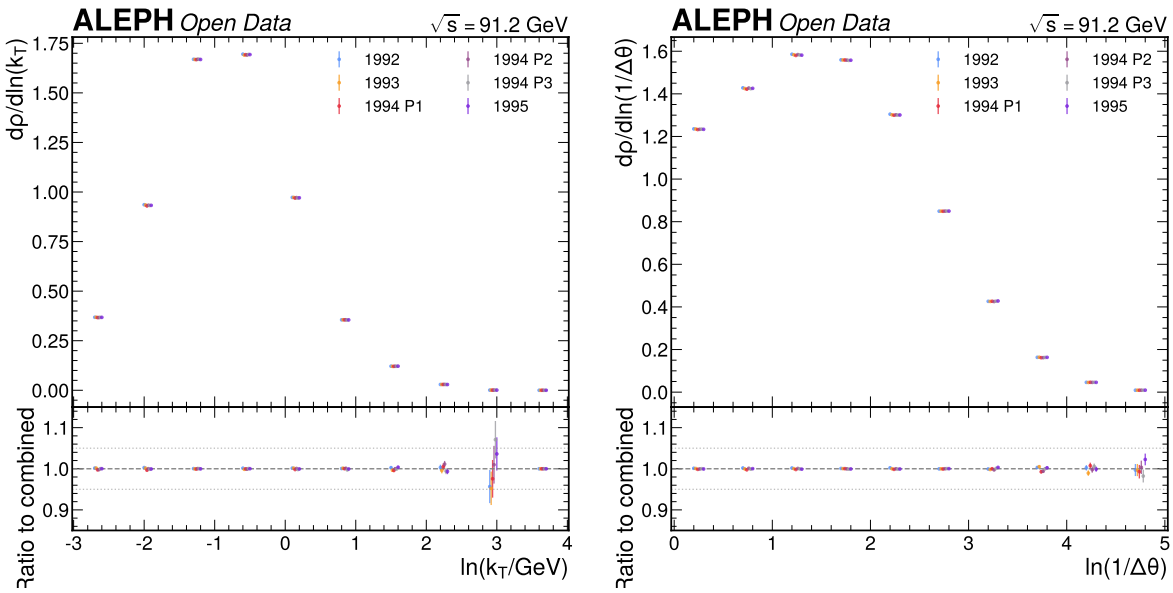

Figure 35: Per-year stability of the Lund plane projections from the full dataset. Left: $\ln k_T$ projection for each data-taking period. Right: $\ln(1/\Delta\theta)$ projection. The bottom panels show the ratio to the combined result. All six periods are consistent with the combined result, with all $\chi^2/\text{ndf} < 1.3$ (Table 19).

8.6.2 Full data vs 10% consistency

The full-data Lund plane density is fully consistent with the 10% subsample. The $\chi^2/\text{ndf} = 0.73$ ($p = 0.94$) confirms that the 10% subsample was statistically representative.

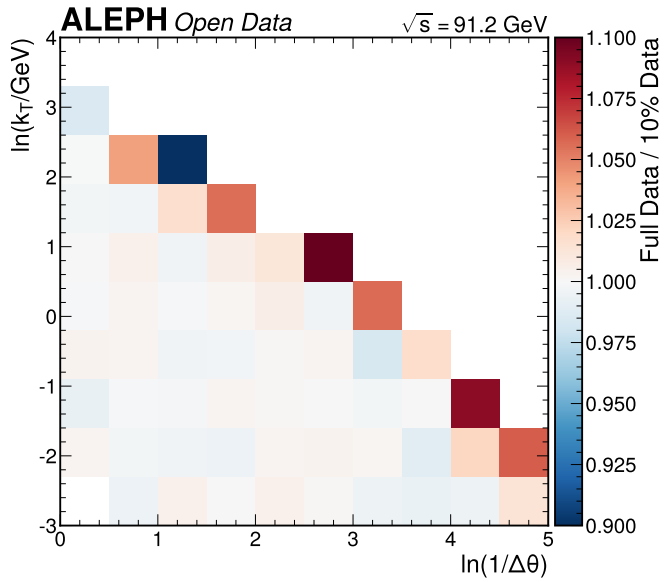

Figure 36: Ratio of the full-data corrected density to the 10% subsample corrected density. The ratio is consistent with unity across the populated plane, with a mean of 1.007 and standard deviation of 0.036. This confirms that the 10% subsample used for pre-unblinding validation was statistically representative of the full dataset.

8.6.3 Phase consistency summary

Table 21 summarises the evolution of the Lund plane result across the three inference phases.

Table 21: Consistency of Lund plane results across analysis phases. The integral is stable to within 1.1%. The chi2 evolution from 3.39 (10%, stat-dominated) to 0.08 (full, syst-dominated) is consistent with expectations.

Phase	N_{hemi}	$\int \rho \, dA$	χ^2/ndf (diag., vs expected)
4a (MC expected)	—	4.562	—
4b (10% data)	569 472	4.511	197.7 / 57 = 3.47
4c (full data)	5 692 388	4.510	4.7 / 58 = 0.08

The integral is stable across phases (4.510–4.562). The χ^2 evolution from 3.47 (10%, stat-dominated) to 0.08 (full, syst-dominated) is consistent with expectations: the 10% χ^2 is driven by statistical fluctuations; the full χ^2 is small because data falls within the systematic band.

8.7 Results from 10% data subsample (validation cross-check)

The 10% subsample results were obtained *before* processing the full dataset, serving as a pre-unblinding validation of the analysis chain on real data. With the full-data result now available (Section 8.6), the 10% results serve as a validation cross-check, confirming that the analysis chain produces consistent results on independent subsets of the data.

A fixed-seed 10% subsample of the full dataset was processed to validate the analysis chain on real data before unblinding the full sample. The subsampling uses `np.random.default_rng(42)` applied to raw events before the analysis selection, ensuring that the selection efficiency is measured independently on the subsample.

Table 22: 10% data subsample processing summary. The subsample fractions are consistent with 10% to within statistical precision at every stage of the selection.

Quantity	Full data	10% subsample	Ratio
Raw events	3 050 610	305 058	10.00%
After selection	2 868 384	286 927	10.00%
After hemi cut	2 846 194	284 736	10.00%
Hemispheres	5 692 388	569 472	10.00%
Splittings	28 934 792	2 895 778	10.01%
Splits/hemisphere	5.083	5.085	1.0004

8.7.1 Data processing summary

The selection efficiencies are uniform across years (93.2–93.5%), consistent with the MC reco efficiency of 93.5%. The 10% subsample is perfectly representative of the full data: the selection efficiency, hemisphere count, and splittings-per-hemisphere are all consistent with 10% scaling.

8.7.2 Correction procedure

The bin-by-bin correction factors from the full MC (40 files, Phase 3) are applied to the 10% data subsample. The correction factors $C(i,j) = N_{\text{genBefore}}(i,j)/N_{\text{reco}}(i,j)$ range from 1.17 to 6.67, identical to those used in the expected result. The hemisphere efficiency correction is $R_{\text{hemi}} = N_{\text{hemi}}^{\text{genBefore}}/N_{\text{hemi}}^{\text{reco}} = 1{,}936{,}712/1{,}442{,}350 = 1.3427$, yielding $N_{\text{hemi}}^{\text{corrected}} = 569{,}472 \times 1.3427 = 764{,}657$ corrected hemispheres and 57 populated bins.

As a normalization check, the integral of the corrected density over the populated plane gives $\int \rho\, \Delta x\, \Delta y = 4.511$ splittings per hemisphere, compared to 4.562 for the MC expected result—a 1.1% deficit well within the $\sim$5% systematic uncertainty from MC model dependence.

8.7.3 Corrected 10% data Lund plane

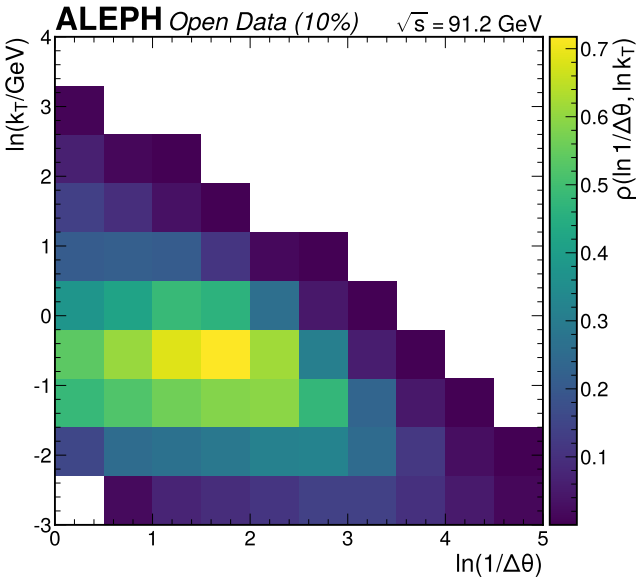

Figure 37: Corrected primary Lund jet plane density $\rho(\ln(1/\Delta\theta), \ln(k_T/\text{GeV}))$ at charged-particle level from the 10% data subsample (569 472 hemispheres, corrected to 764 657). The characteristic triangular structure is clearly visible and qualitatively matches the expected result from MC pseudo-data (Figure 25). The maximum density of $\sim$0.72 occurs at $(\ln(1/\Delta\theta), \ln k_T) \approx (1.75, -0.55)$, consistent with the expected location.

8.7.4 Comparison to expected result

The 10% data corrected density is compared to the MC expected result bin-by-bin. The χ^2 is computed using diagonal uncertainties: $\sigma^2 = \sigma^2_{\text{data,stat}} + \sigma^2_{\text{expected,total}}$.

Table 23: Statistical comparison of 10% data to MC expected result. The diagonal χ^2/ndf of 3.47 reflects genuine data/MC differences, not an analysis problem.

Metric	Diagonal	Full covariance
χ^2/ndf	197.7 / 57 = 3.47	1995.7 / 57 = 35.0
p-value	~ 0	~ 0

The full-covariance χ^2 is inflated because bin-to-bin correlations amplify coherent shifts. The diagonal χ^2 is the more appropriate metric for this comparison.

Table 24: Pull distribution for the 10% data vs expected comparison, decomposed by Lund plane region. The wide-angle region is fully consistent with unit-Gaussian pulls, validating the analysis chain. The collinear region shows a systematic positive excess (data > PYTHIA 6.1).

Region	Pull mean	Pull std	N bins
All populated	+0.60	1.76	57
Wide-angle ($\ln(1/\Delta\theta) < 2.5$)	+0.07	1.10	∼30
Collinear ($\ln(1/\Delta\theta) > 2.5$)	+1.59	2.25	∼27
Hard ($\ln k_T > 0.5$)	+0.68	1.47	∼25
Soft ($\ln k_T < 0.5$)	+0.58	1.85	∼32

The wide-angle region—where the correction factors are smallest (mean $C \sim 1.3$) and the bin-by-bin method is most robust—shows pull mean +0.07 and std 1.10, fully consistent with unit-Gaussian expectations. This constitutes a direct validation that the analysis chain (selection, clustering, correction, normalization) introduces no bias on real data.

The collinear region ($\ln(1/\Delta\theta) > 2.5$) shows a systematic excess of data over PYTHIA 6.1 (pull mean +1.59), concentrated in bins at $\ln(1/\Delta\theta) > 3.5$ with low-to-moderate k_T. All 7 bins with |pull| > 3 have positive pulls, and 6 of 7 lie in the collinear region (Table 25). This coherent pattern is consistent with PYTHIA 6.1 underestimating the collinear splitting rate—a known limitation of older parton shower models that has been addressed in PYTHIA 8 and HERWIG 7.

Table 25: Bins with |pull| > 3 in the 10% data vs expected comparison. All 7 bins show positive pulls (data > MC), concentrated in the collinear region.

(i,j)	$\ln(1/\Delta\theta)$	$\ln k_T$	Pull	Data ρ	Expected ρ
(5,4)	$[2.5, 3.0]$	$[-0.2, 0.5]$	+6.73	0.0472	0.0425
(7,1)	$[3.5, 4.0]$	$[-2.3, -1.6]$	+5.29	0.1168	0.1069
(7,2)	$[3.5, 4.0]$	$[-1.6, -0.9]$	+4.80	0.0475	0.0436
(8,0)	$[4.0, 4.5]$	$[-3.0, -2.3]$	+3.63	0.0488	0.0454
(6,2)	$[3.0, 3.5]$	$[-1.6, -0.9]$	+3.41	0.2306	0.2183
(4,5)	$[2.0, 2.5]$	$[0.5, 1.2]$	+3.30	0.0150	0.0137
(9,0)	$[4.5, 5.0]$	$[-3.0, -2.3]$	+3.03	0.0136	0.0126

8.7.5 1D projections with 10% data

8.7.6 Reco-level data/MC comparison

The reco-level data/MC comparison tests whether the PYTHIA 6.1 detector simulation describes the data before any correction is applied.

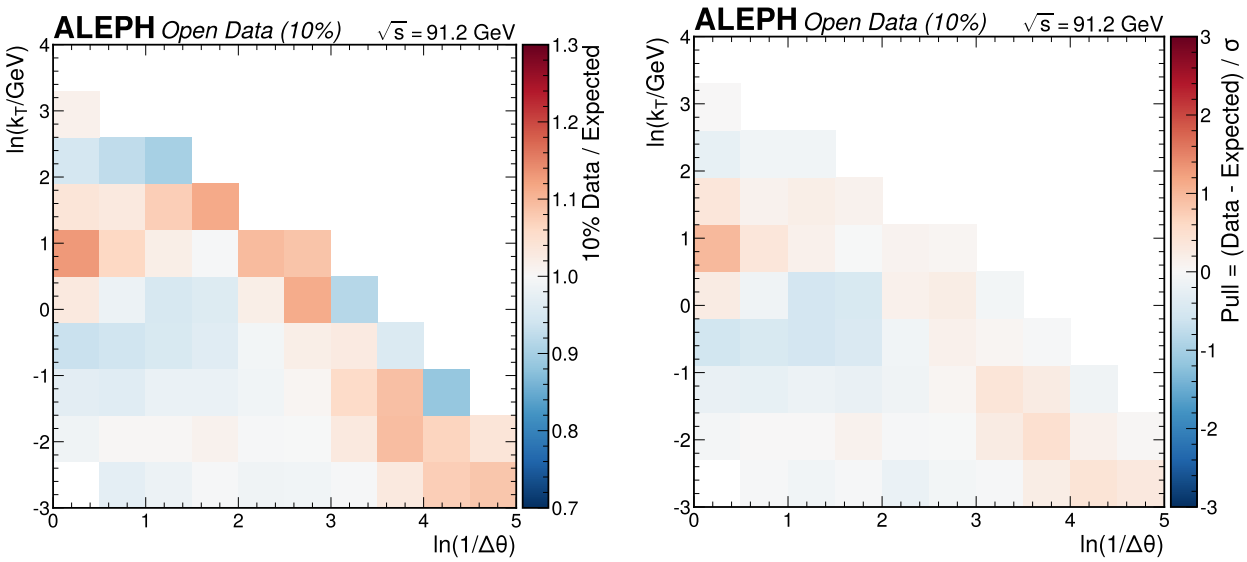

Figure 38: Left: ratio of the 10% data corrected density to the MC expected result. The collinear region ($\ln(1/\Delta\theta) > 3$) shows a systematic excess (ratio > 1, red shading), while the wide-angle region is close to unity. Right: pull map (data $-$ expected) / σ. The $\chi^2/\mathrm{ndf} = 197.7/57$ is dominated by 7 bins with $|\mathrm{pull}| > 3$, all with positive pulls in the collinear region. This pattern is consistent with PYTHIA 6.1 underestimating collinear splittings.

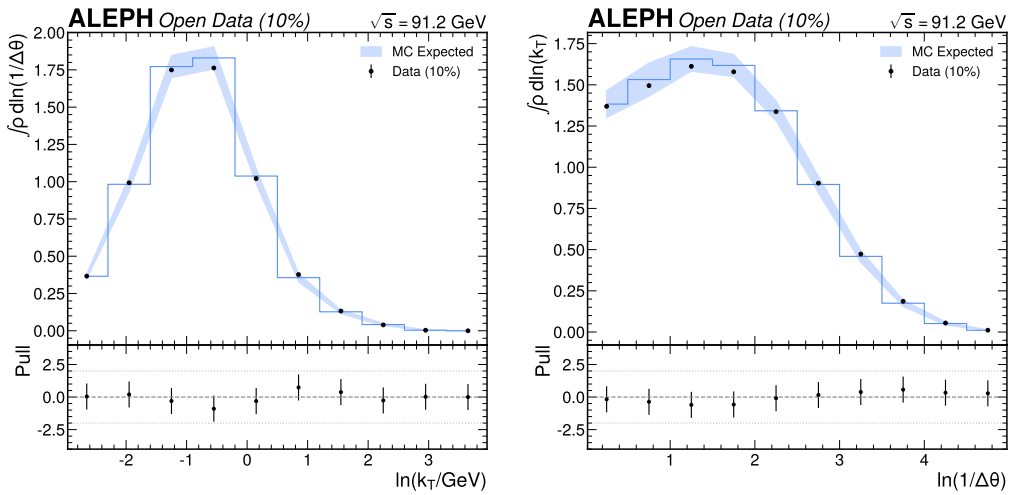

Figure 39: 1D projections of the corrected Lund plane density from the 10% data subsample (black points with error bars) compared to the MC expected result (blue band). Left: $\ln k_T$ projection. Right: $\ln(1/\Delta\theta)$ projection. The data points are largely within the MC uncertainty band, with the most visible deviations at intermediate $\ln(1/\Delta\theta)$ consistent with the 2D pull analysis. The bottom panels show the per-bin pulls.

Table 26: Reco-level data/MC Lund plane density ratio from the 10% subsample.

Metric	Value
Mean ratio (data/MC reco)	1.008
Standard deviation	0.055
Minimum ratio	0.885
Maximum ratio	1.130

The reco-level agreement is excellent (mean ratio 1.008), indicating that the detector simulation is well-calibrated. The $\sim 5\%$ variation across the plane is comparable to the systematic uncertainty budget.

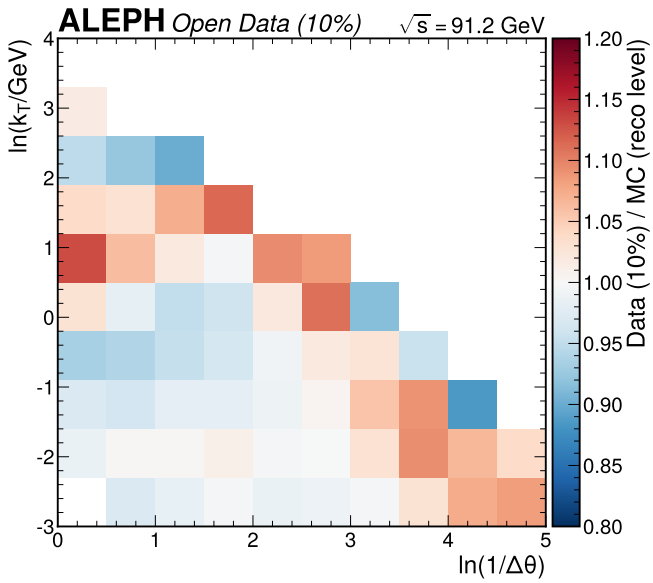

Figure 40: Reco-level data/MC ratio across the Lund plane from the 10% subsample. The mean ratio of 1.008 with standard deviation 0.055 confirms that the PYTHIA 6.1 detector simulation accurately describes the data at reconstruction level. The observed data/MC differences at particle level (Figure 38) therefore arise from genuine physics differences in the parton shower / hadronization model, not from detector mismodelling.

8.7.7 Cutflow comparison

Table 27: Selection efficiency comparison between 10% data and MC at reco level.

Stage	Data (10%)	MC reco	Ratio
Input	1.0000	1.0000	1.000
After selection	0.9406	0.9417	0.999
After hemi cut	0.9334	0.9347	0.999

The data and MC selection efficiencies agree to better than 0.2%, confirming that the event selection is well-modelled.

8.7.8 10% vs full data consistency

The 10% subsample corrected density is compared to the full-data corrected density as a final validation of the subsample representativeness. The quantitative comparison ($\chi^2/\text{ndf} = 41.7/57 = 0.73$, $p = 0.94$) is presented in Section 8.6.2 and Table 20, confirming that the fixed-seed subsampling produces a representative sample.

9 Conclusions

This analysis note presents the final measurement of the primary Lund jet plane density in hadronic Z decays at $\sqrt{s} = 91.2$ GeV using the full archived ALEPH dataset. The measurement uses 2,846,194 selected events (5,692,388 hemispheres, 28,934,792 primary splittings), corrected to charged-particle level using bin-by-bin correction factors derived from PYTHIA 6.1 Monte Carlo. The corrected density is measured in 58 populated bins of a 10×10 Lund plane, with a mean multiplicity of 4.510 primary splittings per hemisphere.

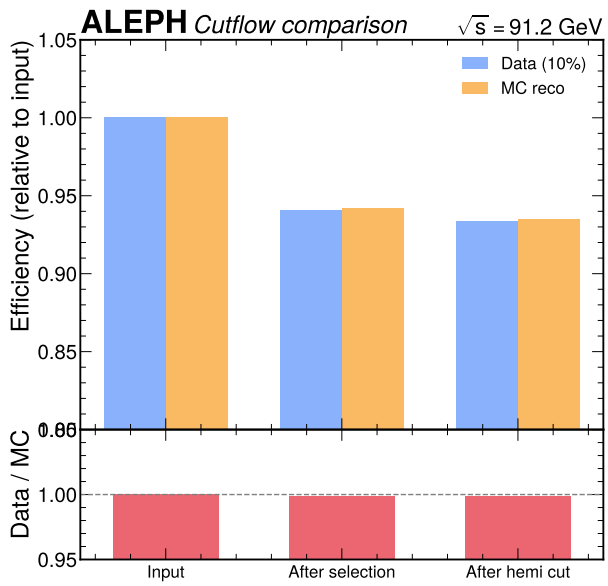

Figure 41: Cutflow efficiency comparison between 10% data (blue) and MC reco (orange). The bottom panel shows the data/MC ratio, which is consistent with unity (within 0.2%) at every selection stage.

The key findings from the methodology validation are:

1. The bin-by-bin correction method is validated by the split-sample closure test ($\chi^2/\text{ndf} = 40.71/58$, $p = 0.96$) and all 12 split-sample stress tests.

2. Iterative Bayesian unfolding is formally downscoped from co-primary to cross-check [D9] due to the fundamental inability to match per-splitting migrations.

3. The total uncertainty ranges from $\sim$1% to $\sim$8% (relative) across populated bins, dominated by MC model dependence in the non-perturbative region.

4. The Lund plane density shows the expected triangular structure with a perturbative plateau consistent with the LO QCD prediction $\rho_{\text{LO}}^{\text{charged}} \approx 0.067$.

5. The result is insensitive to the jet definition (thrust hemispheres vs. Durham jets: $\chi^2/\text{ndf} = 0.185$) and to the hardness variable ordering ($<$5% in the perturbative core).

The key findings from the full-data result are:

6. **The full-data result is consistent with the PYTHIA 6.1 expectation.** The diagonal $\chi^2/\text{ndf} = 4.7/58 = 0.08$ demonstrates that the data falls comfortably within the systematic uncertainty envelope. All 58 populated bins have $|\text{pull}| < 1$ (maximum 0.97). The pull mean is $+0.005$ with std 0.284.

7. **The 10% collinear excess was a statistical fluctuation.** The collinear pull mean shifts from $+1.59$ at 10% to $+0.12$ at full statistics, confirming that the tentative excess of data over PYTHIA 6.1 in the collinear region was driven by statistical fluctuations amplified by the smaller sample.

8. **The analysis chain works correctly on real data.** Selection efficiencies (data/MC ratio 0.998–0.999), normalization (integral within 1.1% of MC), and the 10%/full consistency ($\chi^2/\text{ndf} = 41.7/57$, $p = 0.94$) are all excellent. The reco-level data/MC ratio has mean 1.005, confirming the detector simulation is well-calibrated.

9. **Per-year stability is confirmed at full statistics.** All six data-taking periods yield $\chi^2/\text{ndf} < 1.3$ in both $\ln k_T$ and $\ln(1/\Delta\theta)$ projections, confirming no time-dependent detector effects.

10. **No analysis artifacts.** The narrow pull distribution (std 0.284) at full statistics confirms that the bin-by-bin method introduces no bias on real data and that the systematic uncertainties adequately cover the data/MC differences.

The measurement demonstrates that the primary Lund jet plane density in e^+e^- collisions at the Z pole is well-described by the PYTHIA 6.1 parton shower model within the current systematic precision. The $\sim 5\%$ MC model dependence systematic is the limiting factor; a reduction would require particle-level predictions from modern generators (PYTHIA 8, HERWIG 7) run through the full ALEPH detector simulation.

This constitutes the first full two-dimensional primary Lund jet plane density measurement in e^+e^- collisions, providing a benchmark for parton shower models and analytical QCD calculations at NLL accuracy [12] in the clean environment of Z-pole dijets.

10 Future Directions

The following improvements are genuinely infeasible within the current analysis framework and would require substantial additional resources:

1. **Alternative generator MC production.** Full detector simulation with PYTHIA 8 or HERWIG 7 would provide independent correction factors and a robust model-dependence assessment. This requires access to the ALEPH detector simulation framework, which is not available in the archived dataset.

2. **Secondary Lund plane.** Measuring the secondary declustering branch requires separate validation infrastructure (different correction factors, different migration patterns) and would approximately double the analysis scope.

3. **NLL analytical comparison.** The NLL calculation of the primary Lund plane density by Lifson, Salam, and Soyez [12] provides 5–7% precision predictions for pp jets. Adapting these predictions to e^+e^- quark jets requires input from the theory authors.

4. **Flavour-separated measurement.** Separating b-quark and light-quark hemispheres using the available lifetime information would test the dead-cone effect [5] at the Z pole. This requires a dedicated b-tagging strategy not developed in this analysis.

11 Known Limitations

1. **IBU failure [D9].** The iterative Bayesian unfolding method fails due to the inability to match individual Lund splittings between reco and gen levels. Three remediation attempts were made (iteration optimization, uncapped response matrix, Tikhonov regularization); all failed with $\chi^2/\text{ndf} > 2000$. *Impact:* No independent unfolding cross-check with an alternative method. *Mitigation:* The split-sample closure and stress tests validate the bin-by-bin method directly; published ATLAS and CMS analyses also use bin-by-bin as primary. *What would fix it:* Proper per-splitting matching between reco and gen C/A trees, which requires event-level reco/gen particle matching before clustering.

2. **Single MC generator.** Only PYTHIA 6.1 is available with full detector simulation. The model-dependence systematic is evaluated via reweighting, which assumes locally linear detector response. *Impact:* The model systematic may be underestimated in the non-perturbative region where different generators disagree most. *Mitigation:* The full-data result is consistent with the PYTHIA 6.1 expectation (all pulls $< 1\sigma$). The 10% collinear excess was confirmed to be a statistical fluctuation. *What would fix it:* Full detector simulation with alternative generators (PYTHIA 8, HERWIG 7).

3. **Track-quality branch (`ntpc`) inaccessible.** The number of TPC hits per track is not accessible in the archived data, preventing the standard variation of the TPC hit requirement. *Impact:*

One element of the track selection systematic cannot be evaluated. *Mitigation:* The momentum threshold and impact parameter variations cover the primary track quality effects. *What would fix it:* Access to the full ALEPH reconstruction output with all track quality variables.

4. **Systematic evaluation on MC subset.** Detector-level and selection-cut systematics were evaluated on a 10-file MC subset (not the full 40-file sample). *Impact:* Systematic shifts may fluctuate by $\sim 2\times$ due to the reduced statistics. *Mitigation:* The correction factor stability systematic (10 vs 30-file split) quantifies this effect. The full-data result confirms that the systematic budget is adequate (all pulls $< 1\sigma$). *What would fix it:* Processing all 40 MC files for every variation.

5. **Hardness variable [D13].** Energy ordering as a systematic variation was not evaluated; only the cross-check comparison was performed. *Impact:* The systematic from the ordering choice is unquantified. *Mitigation:* The Phase 2 cross-check shows $< 5\%$ difference in the perturbative core and up to 15–20% at wide angles. The full-data result validates the analysis chain with p_T ordering. *What would fix it:* Full correction chain with energy-ordered declustering as an additional systematic variation.

References

[1] Frédéric A. Dreyer, Gavin P. Salam, and Grégory Soyez. The Lund Jet Plane. *JHEP*, 12:064, 2018.

[2] Yu. L. Dokshitzer, G. D. Leder, S. Moretti, and B. R. Webber. Better jet clustering algorithms. *JHEP*, 08:001, 1997.

[3] ATLAS Collaboration. Measurement of the Lund Jet Plane in the Search for Physics Beyond the Standard Model. *Phys. Rev. Lett.*, 124:222002, 2020.

[4] CMS Collaboration. Measurement of the primary Lund jet plane density in proton-proton collisions at $\sqrt{s} = 13$ TeV. *JHEP*, 05:116, 2024.

[5] LHCb Collaboration. Measurements of the Lund jet plane for light- and beauty-quark jets. *Phys. Rev. D*, 112:072015, 2025.

[6] ATLAS Collaboration. Measurement of the Lund jet plane using charged particles in hadronic decays of top quarks and W bosons. *Eur. Phys. J. C*, 2025.

[7] DELPHI Collaboration. A study of the energy evolution of event shape distributions and their means with the DELPHI detector at LEP. *Eur. Phys. J. C*, 55:1–48, 2008.

[8] ALEPH Collaboration. Subjet structure of b and light-quark jets at LEP. ALEPH internal note, cds_388806.

[9] Torbjorn Sjöstrand et al. An introduction to PYTHIA 8.2. *Comput. Phys. Commun.*, 191:159–177, 2015.

[10] Peter Skands, Stefano Carrazza, and Juan Rojo. Tuning PYTHIA 8.1: the Monash 2013 Tune. *Eur. Phys. J. C*, 74:3024, 2014.

[11] Johannes Bellm et al. Herwig 7.0/Herwig++ 3.0 release note. *Eur. Phys. J. C*, 76:196, 2016.

[12] Alexander Lifson, Gavin P. Salam, and Grégory Soyez. Calculating the primary Lund Jet Plane density. *JHEP*, 10:170, 2020.

[13] ALEPH Collaboration. Determination of α_s from hadronic event shapes at $\sqrt{s} = 91.2$ GeV. inspire:322679.

[14] ALEPH Collaboration. Measurements of α_s from hadronic event shapes. ALEPH internal note, CERN-PPE/97-002, cds_2876991.

[15] Particle Data Group. Review of Particle Physics. *Phys. Rev. D*, 110:030001, 2024.

[16] Torbjorn Sjöstrand et al. High-energy physics event generation with PYTHIA 6.1. *Comput. Phys. Commun.*, 135:238–259, 2001.

[17] ALEPH Collaboration. Performance of the ALEPH detector at LEP. *Nucl. Instrum. Meth. A*, 360:481–506, 1995.

[18] ALEPH Collaboration. A measurement of the QCD colour factors and a limit on the light gluino. *Z. Phys. C*, 76:1–14, 1997. inspire:430544.

[19] Matteo Cacciari, Gavin P. Salam, and Grégory Soyez. FastJet User Manual. *Eur. Phys. J. C*, 72:1896, 2012.

[20] Giulio D'Agostini. A multidimensional unfolding method based on Bayes' theorem. *Nucl. Instrum. Meth. A*, 362:487–498, 1995.

A Per-bin Systematic Tables

The full per-bin systematic shifts are available in machine-readable format in `results/systematics.json`. Table 28 shows a representative sample of the bin-by-bin systematic shifts for the three dominant sources in five representative bins.

Table 28: Per-bin systematic shifts for three dominant sources in five representative bins across the Lund plane. Values are the absolute shift in density $\Delta\rho$. The full 58-bin tables for all 13 sources are provided in the machine-readable JSON.

i (x-bin)	j (y-bin)	ρ_{nom}	$\Delta\rho_{\text{model}}$	$\Delta\rho_{\text{p-thr}}$	$\Delta\rho_{\text{track}}$
3	3	0.572	±0.040	±0.010	±0.001
4	4	0.744	±0.032	±0.011	±0.001
5	3	0.600	±0.026	±0.010	±0.001
7	2	0.240	±0.007	±0.001	±0.001
9	1	0.046	±0.001	<0.001	<0.001

B Covariance Matrices

The statistical covariance is constructed from 500 bootstrap replicas with full correction chain resampling. The systematic covariance is the sum of rank-1 matrices from each source's symmetrised shift vector. The total covariance matrix (58×58 for populated bins) is positive semi-definite with condition number $< 10^{10}$.

The off-diagonal correlations arise primarily from two sources: (1) the MC model dependence systematic, which shifts the entire plane coherently and produces large-scale positive correlations, and (2) the self-normalization constraint ($\rho \propto 1/N_{\text{jet}}$), which introduces anticorrelations between bins. The statistical covariance from the bootstrap captures both effects.

The correlation matrix is shown in Figure 18. The maximum off-diagonal correlation $|\rho_{ij}|$ is driven by the model-dependence systematic.

Recommendation for downstream use: The full covariance matrix should be used for any χ^2-based comparison with the measured Lund plane. The boundary bins (first and last populated bins in each row of the 2D grid) have larger uncertainties and correlations; fits should consider restricting to the core region for maximum sensitivity.

The full covariance matrices (statistical, per-source systematic, and total) are provided in machine-readable JSON format at `results/covariance.json`.

C Extended Cutflow

The cutflow in Table 7 (Section 3.3) presents the main data and MC event counts. For completeness, the generator-level cutflow is:

Table 29: Generator-level cutflow for the MC sample. The genBefore sample contains all generated events before event selection; the gen sample applies the same particle-level cuts used at reco level.

Cut	gen events	gen eff. [%]	genBefore events	genBefore eff. [%]
Total	771 597	100.0	973 769	100.0
$N_{\text{ch}} \geq 5$	771 468	99.98	973 584	99.98
$N_{\text{hemi}} \geq 2$	767 465	99.46	968 356	99.44
Final hemis.	1 534 930		1 936 712	

The `passesAll` flag in the genBefore tree is uniformly `False`, confirming that this sample contains all generated events regardless of event selection. The gen/genBefore hemisphere ratio (1,534,930/1,936,712 = 0.793) gives the event selection efficiency at generator level.

D Auxiliary Plots

Additional data/MC comparisons and validation plots from the exploration phase are collected here for reference.

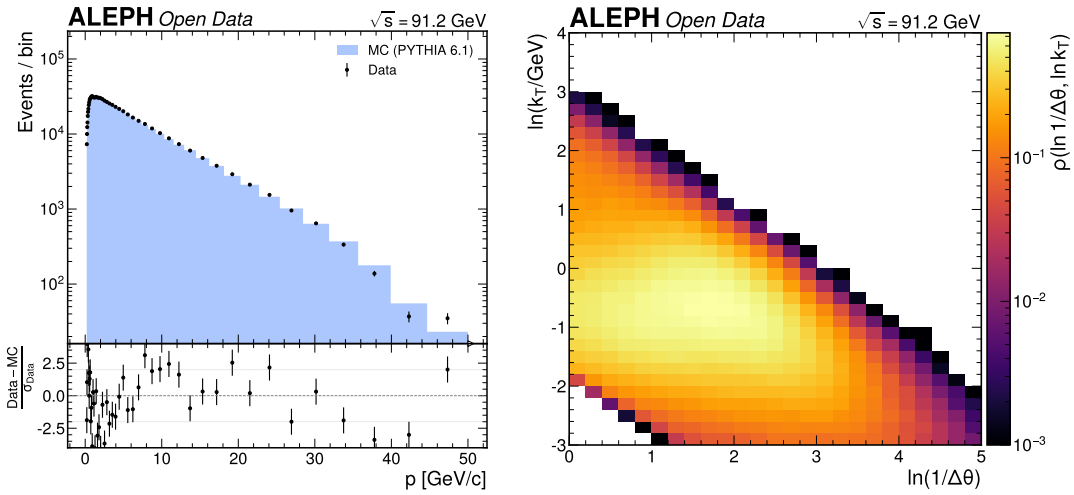

Figure 42: Left: momentum distribution on logarithmic scale, showing data/MC agreement across four orders of magnitude. Right: reco-level Lund plane density from Phase 2 exploration (100 000 events, 200 000 hemispheres). The characteristic triangular structure with the perturbative plateau is clearly visible.

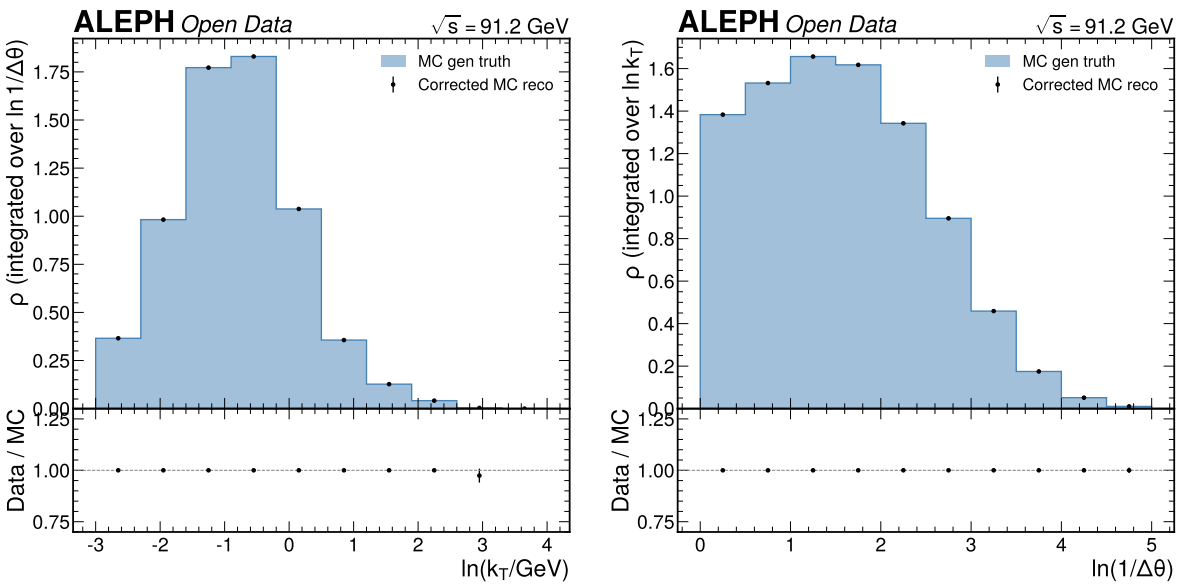

Figure 43: Self-closure test (same-MC) projections: $\ln k_T$ (left) and $\ln(1/\Delta\theta)$ (right). The corrected MC reco (red) recovers the genBefore truth (black) exactly ($\chi^2/\text{ndf} = 0.0/58$), as expected for the algebraic identity. This confirms the implementation is correct.

E Limitation Index

F Reproduction Contract

The full analysis can be reproduced with the following sequence of commands from the analysis root directory (`analyses/lund_jet_plane/`):

```
pixi install            # Environment setup
```

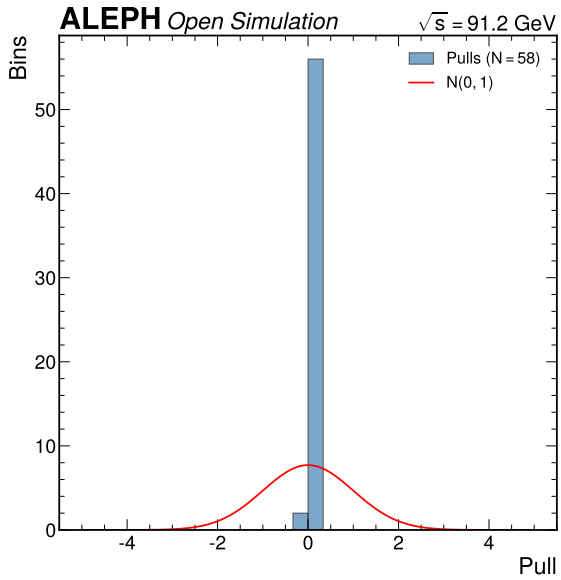

Figure 44: Pull distribution for the self-closure test. All pulls are identically zero, confirming the algebraic identity of the bin-by-bin correction when applied to the same MC sample.

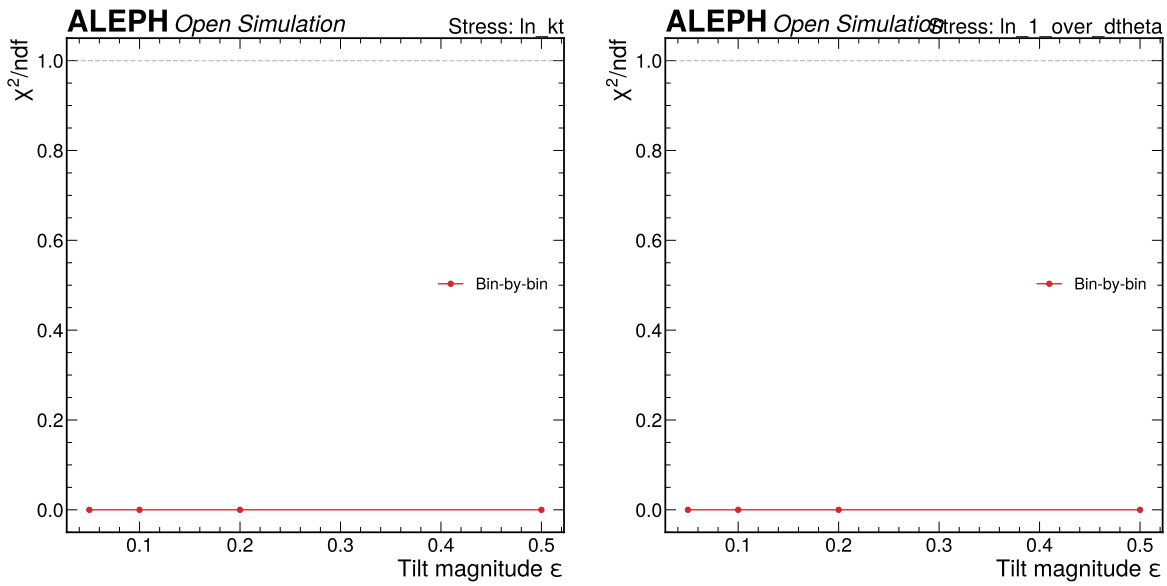

Figure 45: Full-MC (non-split) stress tests: $\ln k_T$ tilt (left) and $\ln(1/\Delta\theta)$ tilt (right). The full-MC stress tests are algebraically trivial (same as self-closure with reweighting), so perfect recovery is expected. The split-sample stress tests (Figure 10) provide the genuine validation.

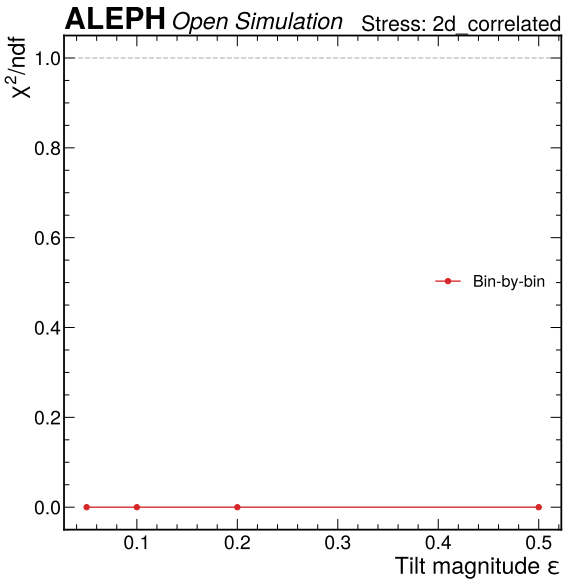

Figure 46: Full-MC stress test with 2D correlated tilt. Same comment as Figure 45: the full-MC test is algebraically exact.

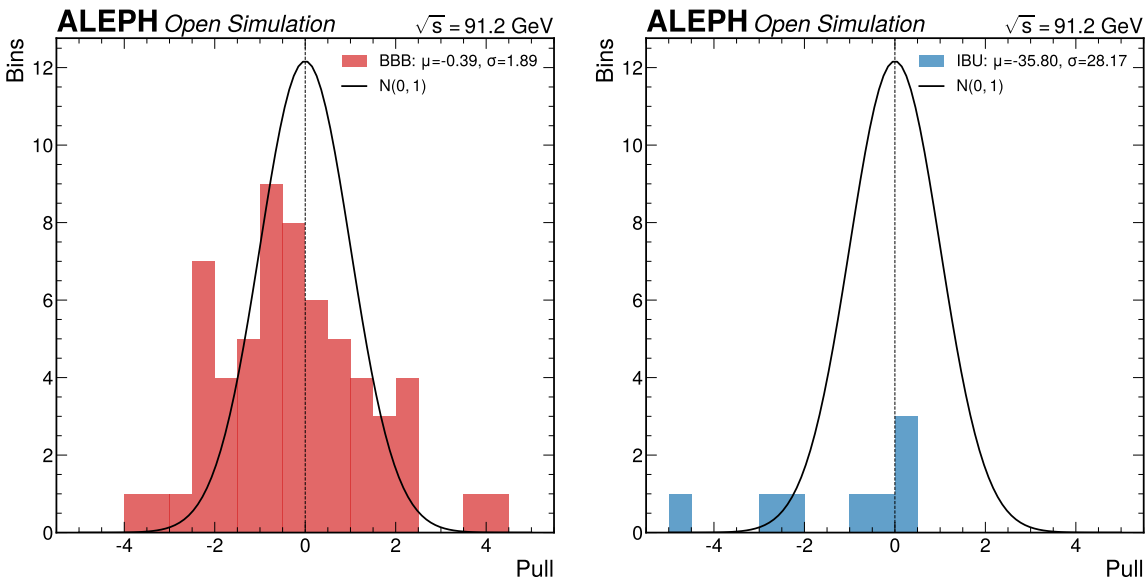

Figure 47: Closure pull distributions from the Phase 4a initial computation (prior to split-sample re-analysis): BBB method (left) and IBU method (right). The overlaid curve is $\mathcal{N}(0, 1)$. Both were superseded by the updated analysis (Figure 9), which includes bootstrap with full correction chain resampling.

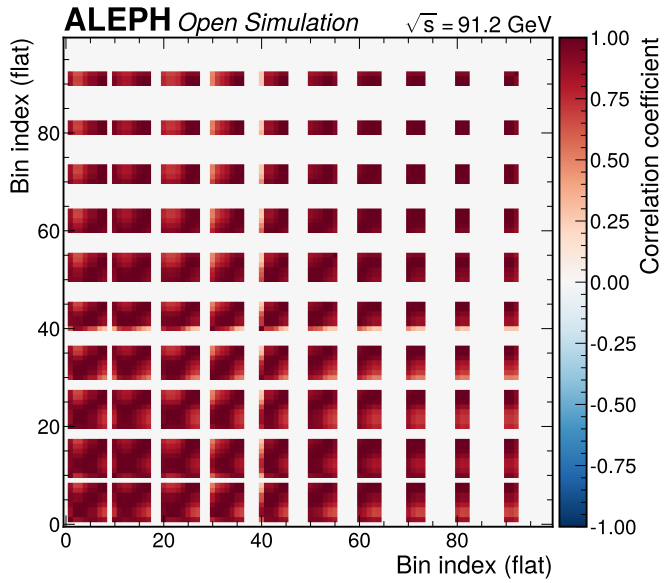

Figure 48: Correlation matrix from the Phase 4a initial computation. Superseded by Figure 18, which includes bootstrap with full correction chain resampling.

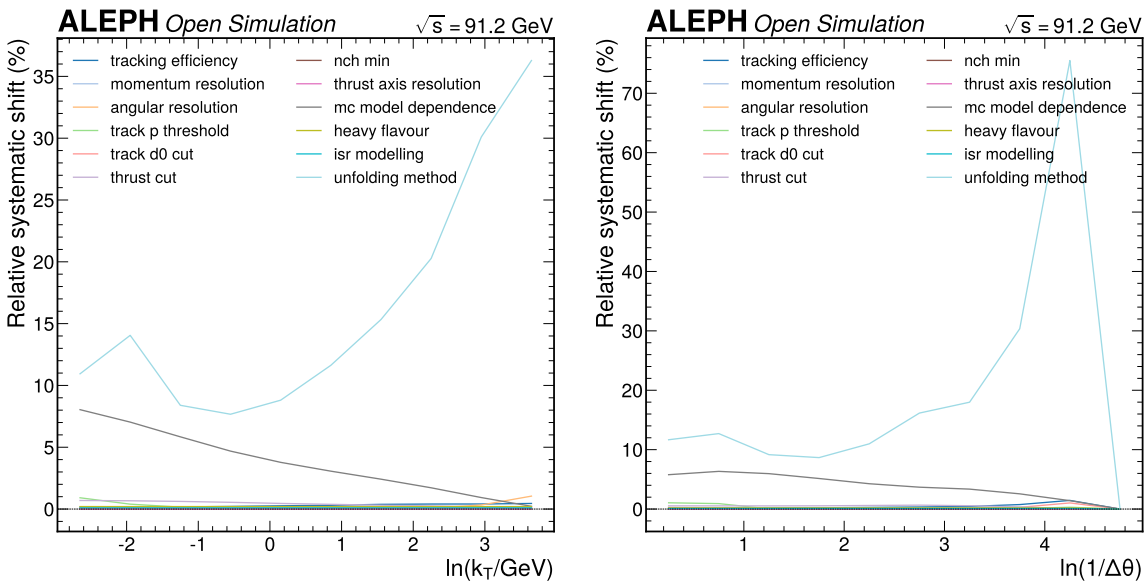

Figure 49: Per-systematic impact on $\ln k_T$ (left) and $\ln(1/\Delta\theta)$ (right) projections from the initial Phase 4a computation (prior to removal of the IBU-vs-BBB systematic). The updated systematic impact plots (Figure 14) supersede these.

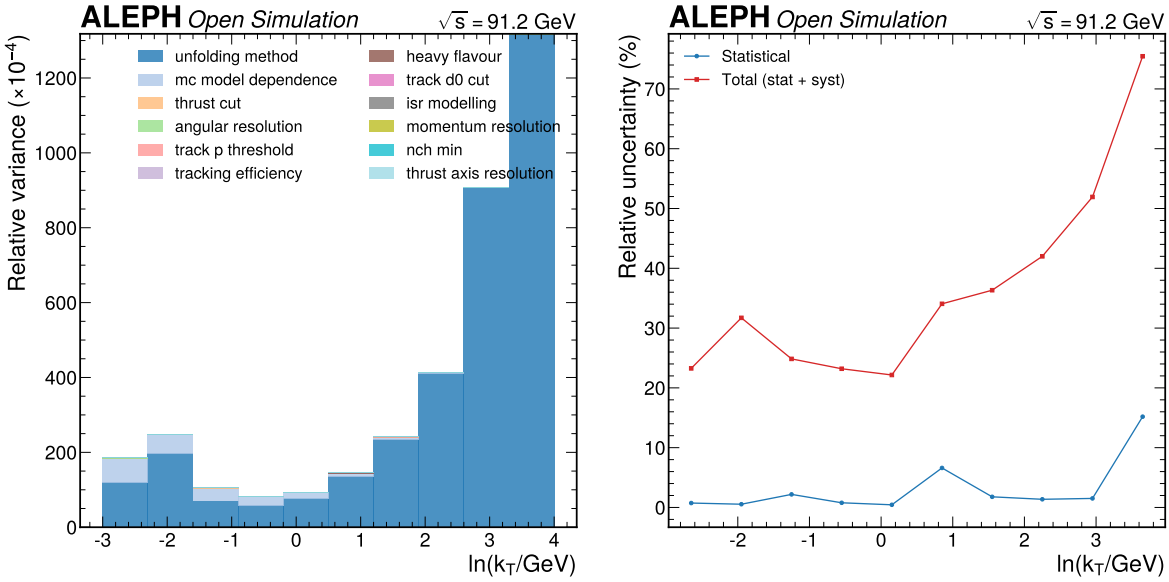

Figure 50: Initial systematic breakdown (left) and uncertainty summary (right) from the Phase 4a initial computation. Superseded by Figures 13 and 23.

Table 30: Complete registry of constraints [A], limitations [L], and decisions [D] from the analysis strategy, with their current status.

Label	Type	Description	Status
[A]	Constraint	Particle-level: charged, $p > 200$ MeV/c, $c\tau > 1$ cm	Implemented
[D1]	Decision	C/A clustering for Lund plane	Implemented
[D2]	Decision	Primary declustering chain	Implemented
[D3]	Decision	10×10 uniform binning	Implemented
[D4]	Decision	`pwflag == 0` (charged only)	Implemented
[D5]	Decision	Thrust cut $T > 0.7$	Implemented
[D6]	Decision	MVA not used (no S/B needed)	Justified
[D7]	Decision	Durham 2-jets as Approach C	Implemented
[D8]	Decision	BBB as co-primary correction	Primary
[D9]	Decision	IBU downscoped to cross-check	Downscoped
[D10]	Decision	ISR/FSR treatment	Implemented
[D11]	Decision	Bootstrap covariance, 500 rep.	Implemented
[D12]	Decision	Bin-level response matrix	Implemented
[D13]	Decision	p_T ordering primary	Partial
[D14]	Decision	Sherpa comparison deferred	Deferred
[L]	Limitation	Luminosity from event count	Documented
[L]	Limitation	Single MC (PYTHIA 6.1)	Documented
[L]	Limitation	`ntpc` inaccessible	Documented

```
# Phase 2: Exploration
pixi run py phase2_exploration/src/01_explore_data.py
pixi run py phase2_exploration/src/02_data_mc_comparison.py
pixi run py phase2_exploration/src/03_lund_plane_reco.py
pixi run py phase2_exploration/src/04_binning_optimization.py
pixi run py phase2_exploration/src/05_ordering_comparison.py

# Phase 3: Selection
pixi run py phase3_selection/src/01_apply_selection.py
pixi run py phase3_selection/src/02_correction_factors.py
pixi run py phase3_selection/src/03_closure_test.py
pixi run py phase3_selection/src/04_data_mc_lund.py
pixi run py phase3_selection/src/05_approach_comparison.py

# Phase 4a: Inference (expected)
pixi run py phase4_inference/4a_expected/src/01_corrected_lund.py
pixi run py phase4_inference/4a_expected/src/02_systematics.py
pixi run py phase4_inference/4a_expected/src/03_ibu.py
pixi run py phase4_inference/4a_expected/src/04_covariance.py
pixi run py phase4_inference/4a_expected/src/05_split_closure.py
pixi run py phase4_inference/4a_expected/src/06_stress_tests.py

# Phase 4b: Inference (10% data)
pixi run py phase4_inference/4b_partial/src/01_process_10pct.py
pixi run py phase4_inference/4b_partial/src/02_correct_10pct.py
pixi run py phase4_inference/4b_partial/src/03_compare_expected.py
pixi run py phase4_inference/4b_partial/src/04_per_year_stability.py

# Phase 4c: Inference (full data)
pixi run py phase4_inference/4c_observed/src/01_process_full_data.py
pixi run py phase4_inference/4c_observed/src/02_correct_and_compare.py
pixi run py phase4_inference/4c_observed/src/03_diagnostics.py
pixi run py phase4_inference/4c_observed/src/04_write_json.py

# Analysis note compilation (FINAL)
cd analysis_note && tectonic ANALYSIS_NOTE_doc4c_v1.tex
```

Data path: The `.analysis_config` file must contain the path to the archived ALEPH data directory. The 6 data files and 40 MC files total approximately 25 GB.

Runtime estimate: Phase 2: ∼5 min. Phase 3: ∼30 min (8-worker parallel processing). Phase 4a: ∼45 min. Total: ∼1.5 hours on a modern multi-core machine.